Nationalization and Firm Performance in Allende's Chile*

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Abstract

We study a large nationalization program using newly assembled firm data from Chile under Salvador Allende (1970–73). Difference-in-differences estimates show that firms targeted by the program experienced a substantial decline in profitability relative to firms that remained private. Return on assets fell sharply, and targeted firms became less likely to report importing activity. Profitability losses were concentrated in manufacturing, where transfers of control were especially contested, while more muted trajectories in strategic and natural-resource sectors were consistent with the state's greater prior involvement and administrative experience in these activities. Additional evidence documents a narrowing public–private wage differential and suggestive declines in electoral support for the governing coalition in more exposed municipalities. These findings suggest that nationalization may be less disruptive when it builds on pre-existing state capabilities and proceeds with broader political and social agreement.

KEYWORDS: nationalization, firms, state
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I. Introduction

The ownership and management of firms is a prominent way in which the state intervenes in the economy (Atkinson and Stiglitz, 1980). Governments nationalize firms to address natural monopolies, coordinate investment, stabilize economies following emergencies, pursue distributive objectives, and obtain control over industries considered strategically important. Examples range from the nationalization of firms in France and the United Kingdom after World War II, to the public takeover of oil and other natural resources in Latin America, and to the broad expansion of state ownership in Eastern Europe and the Soviet bloc.[1] Nationalization is not confined to the past: state ownership expanded during the first two decades of the twenty-first century, particularly following the global financial crisis of 2008 and the COVID-19 pandemic (EBRD, 2020; Megginson et al., 2025). Despite the historical and contemporary importance of these policies, systematic empirical evidence on the consequences of large nationalization programs remains scarce.[2]

We study one of the most ambitious nationalization programs implemented in an electoral democracy. Salvador Allende and the left-wing coalition known as Popular Unity (UP) governed Chile between November 1970 and September 1973 with the objective of pursuing what they called a "democratic transition toward socialism." Nationalization stood at the center of this agenda, alongside higher wages, price controls, intensified land reform, and greater state control over banks and large firms considered strategic to development (Popular Unity, 1969). Through constitutional reform, negotiated acquisitions, public purchases of shares, requisitions, and administrative interventions, the government rapidly expanded the area of social ownership. By 1973, state-owned enterprises accounted for nearly 40 percent of GDP (Hachette, 2000). The process was not uniform: some firms were acquired through negotiated purchases or share transactions, while others were placed under state administration without a settled transfer of ownership. Nationalization therefore unfolded amid labor disputes, owner resistance, restrictions on foreign credit and imported inputs, severe macroeconomic instability, and increasing political polarization.

We assemble a new panel of 71 large Chilean firms observed in 1967-1973 using annual reports submitted to the national securities regulator. We digitize balance sheets and income statements and read approximately 400 reports to extract information on importing, exporting, banking rela-

[1] France nationalized electricity and automobile companies in 1945–1946, while the United Kingdom pursued large-scale nationalization of railways, coal, steel, electricity, and gas under Clement Attlee and the Labour Party. In Latin America, prominent examples include the nationalization of oil companies in Mexico in 1938 and Venezuela in 1976, as well as more recent takeovers of gas, mining, and key services in Bolivia and Argentina.

[2] We do not evaluate temporary nationalizations implemented as stabilization policies with explicit exit strategies, such as Northern Rock in the United Kingdom in 2008 or AIG in the United States in 2008.

tionships, and business activities. Thirty-two firms were ultimately targeted by the nationalization program. They were initially larger and more internationally connected than firms that remained private, although these differences diminish substantially after accounting for firm size and sector. Our baseline design compares the evolution of firms ultimately targeted by nationalization with that of firms remaining private in the same sector. The estimates capture their differential evolution during the Allende period, including anticipation and disruptions preceding formal takeover. Complementary staggered estimates align treatment with firm-specific takeover dates.

Targeted firms experienced 10 percentage points lower return on assets during the Allende period, slightly more than half of the baseline mean, and became less likely to report importing activity. The decline in profitability was concentrated in manufacturing, where the estimated reduction in return on assets was approximately 16 percentage points. Firms in strategic industries and natural-resource activities followed more muted trajectories. The historical record suggests one possible interpretation. Manufacturing was the principal arena of conflict, with interventions and requisitions often following labor disputes and factory occupations while ownership and governance remained unsettled. In several strategic and resource-related activities, the government could instead build on prior state participation, sector-specific personnel, and experience with public financing and coordination. The estimates do not isolate administrative capacity or political conflict from other sectoral differences, but they indicate that outcomes varied with the process through which control was transformed. In addition, firms dependent on foreign exchange, imported inputs, technical licenses, external finance, or established suppliers were vulnerable both to domestic policy changes and to the deterioration of Chile's relations with the United States. We use annual bilateral trade data and synthetic-control comparisons to show that U.S. imports from Chile and U.S. exports to Chile both fell sharply relative to their synthetic counterparts after 1970.

Finally, contemporary surveys show that the wage differential associated with public-firm employment declined during the Allende period, even after accounting for worker characteristics, occupation, and economic activity. Also, municipalities more exposed to plants owned by targeted firms experienced a relative decline in electoral support for Socialist and Communist candidates by 1973. Because workers cannot be linked to particular firms and plant location was not randomly assigned, we interpret these results as changes in the public-private wage differential and suggestive evidence of differential electoral change arising from the nationalization program.

This work contributes first to the empirical literature on nationalization.[3] Governments are

[3]For example, see Stroebel and Van Benthem (2013); Chang et al. (2018). Studies have shown how expropriations arise in equilibrium (Tobin, 1974; Thomas and Worrall, 1994; Bohn and Deacon, 2000; Engel and Fischer, 2010).

more likely to take over foreign-owned companies when commodity prices are high and institutions are weak, and expropriations can reduce productivity and foreign direct investment (Guriev et al., 2011; Melek, 2020; Lucke and Rehfeldt, 2023). Panel data evidence remains limited. Recent exceptions study selection into nationalization and performance in China (Huang et al., 2021; Zhan, 2023) and Russia (Chernykh, 2011).[4] We instead study a country-wide program initiated by an election (Girardi and Bowles, 2018), document selection into treatment, and distinguish the broader Allende-period evolution of targeted firms from changes beginning with formal takeover.

Our work also relates to research on ownership structure and firm performance. Most of this literature studies privatization, and its prominence from the late 1970s onward (Vickers and Yarrow, 1997) produced evidence of substantial productivity advantages for private firms (Megginson et al., 1994; Megginson and Netter, 2001) and motivated research designs based on ownership reforms (La Porta and López-de-Silanes, 1999). Explanations emphasize political incentives in state-owned firms and the gains from "depoliticization" (Shleifer and Vishny, 1994; Barberis et al., 1996; Shleifer, 1998). More recent work shows that outcomes also depend on how ownership transfers are implemented and that irregular or politically connected privatizations can impose costs (Fisman and Wang, 2014; González et al., 2020).[5] Our evidence extends this implementation perspective to nationalization: the expansion of public ownership was not a uniform treatment, and its consequences varied with legal arrangements, prior public capabilities, and political conflict.

Finally, our analysis connects nationalization to the organizational foundations of firm performance and elections. Business-history research treats foreign direct investment as an organizational arrangement through which owners exercise control and provide technical capabilities (Wilkins, 1988). Managers and transferable management practices can have persistent effects on firm performance (Bertrand and Schoar, 2003; Giorcelli, 2019), while Chilean firms were embedded in domestic business groups and networks of banks, suppliers, licensors, and trading partners (Rojas, 2015). Expropriation disputes could also mobilize home governments and reshape access to finance, aid, and trade (Maurer, 2013). We cannot identify these mechanisms separately, but the sectoral and importing results are consistent with their importance. And last, our electoral analysis provides novel evidence consistent with previous work showing that large public policies affect the political equilibrium (Clinton and Sances, 2018; Aidt et al., 2024; Voigtländer and Voth, 2026).

[4]Other empirical studies examine stock-market reactions to nationalizations in France and the United States (Langohr and Viallet, 1986; Ang and Boyer, 2011).

[5]A related literature compares private and state-owned firms after adjusting for observable characteristics; see, for example, DeWenter and Malatesta (2001); Borisova et al. (2015); De Haas et al. (2025).

II. Historical Setting and the Nationalization Program

By the late 1960s, Chile combined rising economic concentration with an increasingly mobilized left (González, 2013). Salvador Allende, leader of the Socialist Party and candidate of the UP coalition, won the September 1970 presidential election with 36.6 percent of the vote.[6] Because no candidate obtained an absolute majority, Congress had to confirm the result, and the following two months were marked by intense polarization and covert U.S. efforts to prevent Allende's inauguration (Kornbluth, 2003; Qureshi, 2009). Congress confirmed Allende after the assassination of General René Schneider strengthened support for constitutional continuity, and he took office on November 4, 1970. The UP sought a "democratic road to socialism" that combined institutional continuity with economic transformation (Popular Unity, 1969). Allende placed the creation of an area of social ownership at the center of this project, declaring that "There can be no socialism without an area of social ownership" (Barros, 2013, pp. 27–28). The associated *Vuskovic plan* combined intensified agrarian reform, nationalization of banks and strategic firms, price controls, and large wage increases (Larrain and Meller, 1991; Girardi and Bowles, 2018). Its diagnosis emphasized concentrated markets, monopoly power, and widespread "excess capacity," while nationalization and demand expansion were intended to raise production, coordinate investment, and redirect surpluses toward social goals (Dornbusch and Edwards, 1990; Edwards, 2023).

The strategy initially produced growth and lower unemployment as higher demand drew on existing capacity, but imports rose and the external balance deteriorated. As reserves declined, the government replaced the fixed exchange rate with multiple official rates and separate import lists. By 1973, ten official rates coexisted and foreign-exchange shortages impeded imports of spare parts. Price controls simultaneously limited firms' ability to adjust output prices, contributing to shortages and black markets (Edwards, 2024, p. 393). The deterioration intensified in 1972 and 1973, when fiscal deficits widened, inflation accelerated, GDP contracted, and real wages fell. By 1973, inflation had reached 441 percent and fiscal deficit 25 percent of GDP (Lüders, 1993).

Objectives and scope of nationalization

The UP program divided the economy into social, mixed, and private areas rather than proposing universal nationalization. The social area would combine existing public enterprises with large-scale mining, financial institutions, foreign trade, major distributors, strategic industrial monopo-

[6]More details about the Salvador Allende government can be found in de Vylder (1976); Boorstein (1977); Sigmund (1977); Corvalán (2003); Harmer (2011); Fermandois (2013); Amoros (2013); Mansuy (2023), among others.

lies, and infrastructure considered essential to development. Small and medium-sized firms were expected to remain predominantly private, and the program promised to protect small shareholders (Popular Unity, 1969).[7] Nationalization was intended to weaken domestic monopoly power and foreign dependence, reshape the productive structure, and place credit, investment, and economic surpluses under greater public coordination. Control of banking was central because it would allow the government to direct credit toward the social area and its investment priorities. The objective was therefore not simply to alter the legal ownership of individual companies, but to reorganize the relationships through which production, finance, investment, and distribution were coordinated.

The best-known component was the 1971 nationalization of copper, which eliminated the remaining U.S. stakes in the major mines and became a focal point of conflict with the affected companies and the U.S. government (Sigmund, 1974; Qureshi, 2009; Edwards, 2025).[8] The broader expansion was unprecedented: state-owned firms accounted for about 14 percent of GDP in 1965 and 39 percent by 1973, when Chile's Production Development Corporation (*Corporación de Fomento de la Producción*, CORFO) controlled more than 500 companies (Hachette, 2000). Public control extended across manufacturing, banking, services, utilities, and natural resources, reshaping relationships among firms, banks, business groups, workers, and public agencies (Larrain and Meller, 1991; Rojas, 2015; Aldunate et al., 2020). It also had major international repercussions. The Nixon administration suspended bilateral assistance and opposed new multilateral lending, while commercial banks reduced their exposure to Chile (Sigmund, 1974; Qureshi, 2009). Access to finance and imported inputs consequently became more difficult while the government was attempting to expand and coordinate the social area. Although historians disagree about the relative contribution of external pressure and domestic policy to the crisis, nationalization became a central axis of conflict within Chile and in its relations with the United States.

[7] Online Appendix A1 reproduces the outline of the nationalization strategy contained in the UP program.

[8] The copper nationalization principally affected the Chilean operations of companies such as Kennecott Copper Corporation and Anaconda Copper Company. These operations do not appear in our firm-level dataset because they did not submit the domestic corporate reports on which our panel is based. Our sample instead includes Chilean-incorporated mining companies subject to the reporting requirements of the Superintendencia de Valores y Seguros. The copper nationalization is therefore important for understanding the historical setting and international response, but the large copper mines do not drive our firm-level estimates.

Implementation across sectors

The expansion of the social area did not follow a uniform procedure.[9] de Vylder (1976, p. 135) distinguishes five principal routes through which the state obtained control of private firms. Expropriation, negotiated purchases, and public acquisition of shares transferred ownership, whereas requisition and intervention transferred administration without formally changing ownership. The instrument depended on firm size and ownership, reliance on foreign technology and imported inputs, owners' willingness to negotiate, and worker organization. Instruments could also be used sequentially, as when intervention was followed by negotiations or an effort to acquire shares. Once a comprehensive statutory framework stalled in Congress, formally temporary interventions and requisitions often became prolonged arrangements. Nationalization was therefore not a single legal treatment imposed at a common date, but a firm-specific process in which authority, ownership, compensation, and permanent governance could remain unsettled.

Manufacturing became the principal arena of conflict, with textiles providing the earliest and most prominent example. Bellavista Tomé was expropriated after suspending operations, while Yarur was intervened following a major labor conflict in April 1971. The government subsequently extended control through expropriations, requisitions, and interventions, often relying on DFL 520, a 1932 decree permitting action when producers of essential goods interrupted production, left capacity idle, or violated price regulations. Strikes and factory occupations frequently preceded intervention and sometimes exceeded the government's plans. Barros (2013, pp. 169–260) reconstructs this sequence, while Fermandois (2013, pp. 384–391) places it within the formation of the social area. Ownership, compensation, and permanent governance often remained unresolved; owners contested the transfers through courts, parties, business associations, and the press, while workers and parties within the UP pressed to retain intervened firms. The textile takeovers consequently became both changes in administration and a turning point in organized business resistance (Silva, 1996, ch. 3). Similar variation appears among firms in our dataset. A Ministry of Economy document recommended maintaining the requisition of Compañía Cervecerías Unidas while seeking its shares, negotiating with the Swedish owners of Compañía Chilena de Fósforos, and completing an agreement with Compañía Chilena de Navegación Interoceánica (de Vylder, 1976, p. 142). Formal takeover dates therefore need not mark the beginning of disruption: negotiations, labor mobilization, threats of intervention, and uncertainty could affect operations beforehand.

Strategic and natural-resource activities often entered the period with greater prior public in-

[9]Online Appendix A2 provides additional detail on the legislative proposals, legal instruments, and political disputes surrounding the formation of the social area.

volvement. Successive governments had used CORFO to establish enterprises, finance investment, provide technical assistance, and participate in mixed companies. Under Eduardo Frei, the "Chileanization" of copper gave officials experience with joint ownership and sectoral oversight (CORFO, 2009, pp. 105–106). In fishing, CORFO had financed vessels and infrastructure and coordinated the integration of 21 firms into six industrial groups, including Coloso and Indo, following the anchoveta crisis; it supplied approximately 46 percent of the capital committed to the restructuring (CORFO, 2009, pp. 126–133). Public control in such sectors could build on existing ownership stakes, specialized personnel, and administrative routines. This was not a sharp division between orderly nationalization in strategic sectors and improvised intervention in manufacturing. Copper enjoyed broad domestic support but generated a severe international compensation dispute (Fermandois, 2013, pp. 435–440), while some industrial firms entered the social area through negotiation. Nevertheless, sectors differed in prior state involvement, legal preparation, worker mobilization, owner resistance, and foreign dependence. Barros (2013, pp. 137–139) notes that incorporating nitrate, iron, coal, steel, and cement generated less domestic conflict than the textile interventions. These differences provide historical grounds for the heterogeneous effects examined below. On September 11, 1973, the armed forces overthrew Allende and initiated a seventeen-year dictatorship (Huneeus, 2006); the military government subsequently reversed many nationalizations as part of a wider shift toward market-oriented policies.

III. Data and Sample Construction

Firm reports and reporting population

We construct a panel dataset from business reports collected from public archives and state libraries. During the period we study, firms listed on the Chilean stock market and other widely held corporations were legally required to submit annual reports to the *Superintendencia de Valores y Seguros* (SVS), a regulatory agency comparable to the U.S. Securities and Exchange Commission. The requirement covered firms with dispersed ownership, including companies with more than five hundred shareholders. The reports therefore originate from a contemporaneous regulatory obligation rather than from a retrospective selection of firms based on their later nationalization status or economic performance. At the same time, the SVS population was not a random sample of all Chilean businesses: it disproportionately represented large, formally incorporated corporations with dispersed ownership and excluded many small firms and informally organized producers.

The surviving records nevertheless cover many of the country's most economically important

enterprises.[10] In 1969, firms in our dataset accounted for almost two-thirds of electricity generated for sale to third parties and nearly half of the country's maritime transport capacity. A single firm produced more than half of national wood-pulp output, while the three fishing companies in the sample produced roughly one-third of Chile's fish meal; wood pulp and fish meal represented 23 and 14 percent, respectively, of the value of non-mining exports in that year (Alaluf, 1971; Larraín and Meller, 1991). SVS reports have also been used extensively in previous research on Chilean firms, business groups, and corporate governance (Khanna and Palepu, 2000; Martínez et al., 2007; González and Prem, 2018, 2020; González et al., 2020; Aldunate et al., 2020). Our results should consequently be interpreted as applying to large, regulated corporations operating in economically important sectors rather than to the full distribution of Chilean firms.

Construction of variables

The reports combine financial statements with descriptions of firms' activities. We digitize assets, fixed assets, debt, earnings before interest and taxes, and sales from balance sheets and income statements. Although their presentation varies somewhat across firms and years, the accounting information is sufficiently standardized to construct comparable measures of return on assets, sales over assets, leverage, and the composition of assets and liabilities. We also read approximately 400 reports to extract information from the discussions addressed to shareholders. These "business letters" describe firms' principal activities and the conditions affecting their operations, allowing us to code whether a firm reports importing inputs or exporting products in each year between 1967 and 1973 and to identify relationships with domestic, international, and U.S. banks.

The descriptions of operations also allow us to classify firms into three sectors. Primary-sector firms engage in natural-resource extraction and agriculture; secondary-sector firms in manufacturing and industrial production; and tertiary-sector firms in services, transport, commerce, and utilities. We also separately identify manufacturing firms and construct indicators for strategic and natural-resource activities. The strategic classification follows the activities identified for public control in the UP program, while the natural-resource classification includes firms principally engaged in mining, forestry, fishing, or the direct processing of natural-resource inputs. Finally, we determine whether and when each firm was nationalized using academic articles, historical books, the reports themselves, state records, and the findings of special commissions (Farías, 2000; Cor-

[10]These include major industrial producers such as Compañía Manufacturera de Papeles y Cartones, Cemento El Melón, and Elecmetal; key utilities and infrastructure firms such as Empresa Nacional de Electricidad and Compañía de Gas de Santiago; large transport and shipping companies such as Compañía Sud Americana de Vapores; and prominent food, beverage, and consumer-goods producers such as Compañía Cervecerías Unidas and Carozzi.

valán, 2003; Congress Report, 2004; CORFO, 2009; Barros, 2013; Aguirre-Briones, 2019). We distinguish firms that remained privately controlled from those eventually targeted by the program and record the date on which the government formally assumed control, allowing us to separate the broad Allende-period estimand from the specification aligned with formal takeover.

Estimation sample and panel coverage

We located and digitized usable information for 87 firms. The baseline design requires a firm to be observed at least once before Allende's election and at least once during his administration, so that changes can be estimated within the same firm. Seventy-one firms meet this temporal-support requirement and constitute the estimation sample; the remaining 16 lack observations on both sides of the change in administration. Of these 71 firms, 32 were ultimately targeted by the nationalization program and 39 remained private. Manufacturing accounts for 33 firms, including 22 targeted firms. The remainder span agriculture, forestry and fishing, mining, electricity and gas, transportation, commerce, finance, real estate, and other services. We additionally classify 25 firms as strategic, 14 of which were targeted, and 13 as engaged in natural-resource extraction or processing, 6 of which were targeted. Eleven firms belong to both categories. These smaller and overlapping groups permit sectoral comparisons but yield less precise estimates.[11]

Table A1 compares firm-level averages over 1967–1970 for firms entering the estimation sample and those that do not. The groups are similar in nationalization status, size, leverage, importing activity, sector, geographic presence, and banking relationships. Included firms are more likely to export and have somewhat higher profitability, although the latter is only marginally significant and a joint test does not reject that the observed characteristics have no conditional explanatory power for sample inclusion. The panel is unbalanced because some reports were not preserved and some surviving reports lack particular variables. In the baseline profitability panel, 29 firms are observed in all seven years, 32 for four to six years, and 10 for two to three years, producing 390 firm-year observations. Table 1 reports annual coverage by nationalization status and takeover cohort. The number of observed firms falls from 59–62 per year in 1967–1970 to 49–51 in 1971–1973, but this

[11]The sample includes major firms in paper and cellulose, food and beverages, cement, metal products, fishing, forestry, electricity and gas, and maritime transport. Examples include Compañía Manufacturera de Papeles y Cartones, Compañía Cervecerías Unidas, Elecmetal, the cement producers Polpaico and El Melón, the fishing companies Eperva, Indo, and Coloso, and the shipping companies Compañía Chilena de Navegación Interoceánica and Compañía Sud Americana de Vapores. The large U.S.-owned copper mines and the principal nitrate enterprises are not included because they did not belong to the domestic reporting population from which the panel was constructed. Prominent textile cases such as Yarur and Bellavista Tomé are also not in the estimation sample; we use them to characterize the broader implementation environment in manufacturing, rather than as firm-level cases underlying our estimates.

decline is not concentrated among targeted firms: between 1970 and 1973, the number of private firms falls from 34 to 26, while the number of targeted firms changes only from 25 to 24.

We also examine whether missing observations in the panel are systematically related to nationalization status or pre-existing firm characteristics. Table A2 shows that nationalization status, size, profitability, international exposure, and banking relationships do not significantly predict the number of missing observations either before or during the Allende administration. Together with the annual counts in Table 1, these results provide no evidence that the unbalanced panel reflects the selective disappearance of targeted firms, although the modest number of firms limits precision in analyses by sector or takeover cohort. We end the panel in 1973 because the September 11 coup abruptly terminated the Allende government and initiated the reversal of nationalization, changes in management and labor policy, and a different macroeconomic and political regime. Post-coup observations would therefore combine the legacy of the UP program with regime change and subsequent ownership reversals. We interpret the estimates as short-run outcomes during the program's implementation; although some December 1973 reports include the first months after the coup, the results are similar when all 1973 observations are excluded.

IV. Research Design

Selection into the nationalization program

Table 2 compares firms eventually nationalized with those that remained private using characteristics measured over 1967–1969, the last years before Allende's election. The table contains 66 firms with the required information, including 30 subsequently nationalized firms and 36 that remained private. Firms ultimately nationalized were substantially larger, held more debt, participated more frequently in international markets, maintained relationships with more domestic and international banks, and were more likely to operate in the secondary sector. Much of this imbalance reflects the size and sectoral composition of the firms selected by the UP program. Once column 4 controls for log total assets and sector fixed effects, differences in performance, leverage, international participation, and most banking relationships become smaller and statistically insignificant. The principal remaining imbalance is the number of relationships with U.S. banks, which motivates a robustness exercise allowing firms with such relationships to evolve differently during the Allende period. Among nationalized firms, the characteristics reported in column 5 do not significantly predict the year of takeover, although the small sample limits the precision of this comparison.

Table A3 examines selection more directly using firm-level averages from 1967–1969. Conditional on sector, firm size is a strong and stable predictor of eventual nationalization, consistent with the program's emphasis on large and economically important enterprises. Pre-Allende return on assets and sales over assets do not jointly predict nationalization once size and sector are taken into account, and exporting and importing activity provide little additional predictive power. Balance-sheet characteristics and banking relationships are jointly predictive when considered as a group, largely because firms with more relationships with national banks were less likely to be nationalized, but the additional covariates are not jointly significant in the full specification. Selection into the program was therefore systematic, particularly with respect to firm size and sector, but the available evidence does not indicate that firms were selected because their profitability was already deteriorating. Observable characteristics cannot exclude selection based on political salience, labor conflict, managerial decisions, or expectations about future performance. Our design consequently does not assume random assignment; it compares changes within firms while accounting for contemporaneous shocks shared by firms operating in the same sector.

Baseline Allende-period estimand

We estimate the effects of being targeted by the UP nationalization program using the panel of 71 firms observed between 1967 and 1973. The baseline design compares firms nationalized at some point during the Allende administration with firms that remained privately controlled throughout the period. We begin with a flexible difference-in-differences econometric model:

$$Y_{ijt} = \sum_{k \neq 1969} \beta_k \left(N_i \times T_k\right) + \phi_i + \phi_{jt} + \varepsilon_{ijt} \tag{1}$$

where Y_{ijt} is an outcome for firm i, operating in sector j, in reporting year t. The indicator N_i equals one for firms nationalized during the Allende administration, while T_k identifies each reporting year, with 1969 omitted. Firm fixed effects ϕ_i absorb time-invariant differences across firms, and sector-by-year fixed effects ϕ_{jt} absorb macroeconomic and sectoral shocks shared by firms operating in the same sector and year. Standard errors are clustered at the firm level. The coefficients therefore compare changes among targeted and private firms within sectors rather than attributing aggregate movements during the Allende period to nationalization.

Fiscal-year timing is important for interpreting equation (1). Firms closed their reports either in July or at the end of December.[12] Because Allende was elected in September 1970, the

[12] Among the 71 firms in the sample, 26 reported in July and 45 reported in December.

1970 observation precedes the election for July reporters but includes the first months after the election for December reporters. We omit 1969 because it is the last reporting year that unambiguously precedes the election for every firm. Accordingly, $\widehat{\beta}_{1970}$ is a transition-year estimate rather than a purely pre-treatment or fully post-treatment coefficient; the estimates for 1967 and 1968 provide evidence on earlier differential trends, while those for 1971–1973 capture relative performance during the Allende administration. These coefficients do not isolate the formal transfer of ownership. Instead, they measure the differential evolution following Allende's election of firms ultimately targeted by the program, bundling anticipation, negotiations, political and operational conflict, changes in access to inputs or finance, and formal takeover whenever it occurred. We interpret them as Allende-period intention-to-treat effects.

Identification requires that, absent the nationalization program, targeted firms would have followed trends similar to those of private firms operating in the same sector. Firm fixed effects address permanent differences across companies, while sector-by-year fixed effects absorb contemporaneous shocks common within sectors, but neither eliminates firm-specific, time-varying factors correlated with selection. We also estimate a parsimonious specification summarizing the average differential performance of targeted firms during the Allende period:

$$Y_{ijt} = \beta (N_i \times A_{it}) + \phi_i + \phi_{jt} + \varepsilon_{ijt} \tag{2}$$

where A_{it} is a firm-specific Allende-period indicator based on the fiscal-year closing date. It equals one beginning in 1970 for December reporters, whose 1970 observation includes months following the election, and beginning in 1971 for July reporters, whose 1970 observation closes before it. The coefficient $\widehat{\beta}$ is our baseline estimate of the average Allende-period effect of being targeted by the program. We estimate equations (1) and (2) for return on assets, sales over assets, and indicators for whether a firm reports exporting or importing activity.

Takeover timing and staggered treatment

We also exploit variation in the timing of formal takeover. Although the nationalization program was announced before Allende assumed office, implementation was staggered: some firms were taken over during 1971, while others remained privately managed until 1972. A firm becomes treated in the first fiscal-year report closing after the date on which the government formally assumed control and remains treated thereafter. Event time zero therefore corresponds to the first report covering a period under state control, negative values correspond to reports closing before takeover, and positive values correspond to subsequent reports; firms never nationalized re-

main untreated throughout. We estimate average and dynamic effects using the imputation-based difference-in-differences estimator proposed by Borusyak et al. (2024). The estimator first uses observations from firms untreated in a given year to estimate the untreated outcome process with firm and sector-by-year fixed effects, then imputes the untreated potential outcome for each treated firm-year and aggregates the differences between observed and imputed outcomes. Unlike conventional two-way fixed-effects estimation, it does not use already-treated firms as controls for firms treated later and accommodates treatment effects that vary across firms and over time.

Because the nationalization program was publicly known, anticipation may have affected firms before formal takeover. The post-takeover estimates therefore cannot isolate public ownership from the political and organizational process surrounding the transfer of control. We treat them as complementary to the broader Allende-period estimates rather than as a preferable design.

V. Firm Performance

Baseline Allende-period estimates

Figure 1 presents estimates of equation (1) and summarizes our main firm-level results. Targeted firms experienced a pronounced decline in return on assets, with the largest differences emerging in 1972 and 1973 (panel A). Sales over assets also declined in those years, but the estimates are less precise (panel B). We find no clear differential change in exporting activity (panel C), whereas firms targeted by the program became substantially less likely to report importing activities (panel D).[13] The patterns are similar after controlling for baseline firm size interacted with year fixed effects (Figure A1). The 1970 coefficients require caution because reports closing in July precede Allende's election, while those closing in December include the first months after it. Estimates for profitability, sales, and exporting are close to zero in this transition year, but importing is already lower among targeted firms, consistent with an early adjustment in international business relations. Before the election, we find little evidence of differential trends: joint tests of $\widehat{\beta}_{1967}$ and $\widehat{\beta}_{1968}$ yield p-values of 0.89, 0.14, 0.75, and 0.85 for return on assets, sales over assets, exporting, and importing, respectively. These tests exclude the transition-year observation and compare the groups relative to 1969; Table A4 provides complementary evidence using the pre-election period.

Table 3 reports the average Allende-period estimates from equation (2). Targeted firms expe-

[13]Consistent with the expansion of the Chilean economy in 1971, driven in part by a sharp increase in public spending, we find suggestive but statistically insignificant evidence of higher sales among targeted firms in that year.

rienced a decline in return on assets of approximately 9–10 percentage points, or about half of the baseline mean, and the estimate remains precisely estimated after controlling for baseline firm size. The estimates for sales over assets are also negative, but they are smaller and considerably less precise. Comparing the two outcomes does not identify a particular mechanism: lower accounting returns could reflect lower sales, controlled output prices, higher input or financing costs, production interruptions, changes in product composition, or changes in accounting practices.[14] The available reports do not permit us to distinguish among these possibilities. The exporting estimates are small and statistically imprecise, whereas the decline in reported importing is substantially larger. We therefore interpret the baseline evidence as strongest for accounting profitability, with reduced importing providing supporting evidence of international disengagement. The results for sales over assets and exporting do not support comparably strong conclusions.

Estimates using takeover timing

Figure 2 presents event-study estimates obtained using the imputation-based estimator proposed by Borusyak et al. (2024). A firm becomes treated in the first fiscal-year report closing after its formal takeover date, and event time zero denotes that first report. Because annual reports cover the preceding fiscal year, the observation at event time zero may include months both before and after the transfer of control. The post-takeover estimates broadly resemble the baseline patterns but are less precise. Return on assets declines following takeover, with larger negative estimates at longer horizons, and sales over assets display a similar trajectory. Exporting shows no systematic change, whereas importing declines after takeover, again most clearly at later event times. The wider confidence intervals reflect the limited number of firms, takeover cohorts, and post-treatment observations available once treatment is aligned with firm-specific dates.

The coefficients preceding formal takeover are small and statistically indistinguishable from zero, providing no evidence of differential trends immediately before state control. They should not, however, be interpreted as conventional placebo estimates because the nationalization program was publicly known and firms, workers, creditors, and suppliers could respond once takeover became likely. Such anticipation would appear at negative event times and could attenuate the contrast around formal takeover. The staggered design consequently estimates a narrower object than the baseline specification but still does not isolate public ownership from the political and

[14] Additional balance-sheet results in Table A5 and Figure A2 show no systematic differential change in total assets, fixed assets, or leverage among targeted firms, which reduces concerns that the return-on-assets estimate is driven entirely by a change in its denominator. These patterns do not, however, rule out changes in accounting or reporting practices within particular balance-sheet or income-statement categories.

organizational process accompanying the transfer of control. We interpret it as complementary evidence that the negative patterns in profitability and importing remain visible when treatment follows takeover timing, while the limited number of firms and cohorts constrains precision.

Robustness, inference, and spillovers

Table A8 evaluates alternative specifications and estimators. Collapsing the panel into periods before and during the Allende government addresses concerns about serial correlation (Bertrand et al., 2004), while excluding the 1973 observations of December-reporting firms addresses the fact that these reports include the first months after the September 11 coup. We obtain similar estimates in both exercises. The results are also similar when replacing sector-by-year fixed effects with narrower industry-by-year fixed effects and when allowing firms with pre-Allende links to U.S. banks—the principal remaining imbalance in Table 2—to experience a distinct Allende-period change. The main findings are therefore not driven by observations recorded immediately after the coup, broad sectoral shocks, or the measured imbalance in U.S. banking relationships.

Table A9 shows that initial international exposure does not explain the results. Allowing firms active in international markets in 1969 to follow separate Allende-period trajectories leaves estimates unchanged. Matching exercises in Table A8 likewise indicate that observable baseline differences do not account for the profitability result. Allowing for bounded deviations from parallel trends using recent approaches lead to similar conclusions (Figure A3; Rambachan and Roth, 2023), and Figure A4 shows that no single targeted firm drives the baseline estimates.

Similarly, inference based on a wild cluster bootstrap, reported in Table 3, confirms the statistically significant estimates. In addition, we examine whether private firms in industries heavily exposed to the program provide contaminated comparisons. Table A10 excludes non-nationalized firms from either the single most intensively nationalized industry or the three most intensively nationalized industries, measuring intensity using both the share of firms nationalized and their share of pre-1970 industry assets in the 87-firm archival sample. Profitability estimates remain negative across all specifications and are statistically significant in three of the four exercises, while importing estimates remain negative but become less precise. These exercises indicate that the performance results are not generated by comparison firms in the industries most exposed to nationalization, and they also illustrate the loss of precision from further sample restrictions.

Heterogeneous effects

Table 4 shows that the profitability decline is concentrated among manufacturing firms. This pattern is consistent with the historical record, which describes manufacturing as the principal arena of conflict: "it was in manufacturing industry that the most bitter fights between the Left and the rightist opposition took place after December 1970" (de Vylder, 1976, p. 134). By contrast, we do not find comparable negative profitability estimates among firms classified as operating in strategic sectors, defined following the UP program (Popular Unity, 1969) as activities central to development, economic sovereignty, and planning, or among firms operating in natural-resource industries.[15] These contrasts should be interpreted cautiously because the strategic and natural-resource categories contain relatively few targeted firms, overlap with one another, and yield wide confidence intervals. The estimates therefore indicate different and potentially more muted trajectories outside manufacturing rather than establishing that these sectors were unaffected.

Pre-existing relationships with state-owned banks provide a related dimension of heterogeneity. Table A11 interacts the treatment effect with an indicator for firms connected to a state bank in 1969 and shows that the decline in performance was substantially smaller among firms with such connections. This pattern is consistent with pre-existing financial links to the state mitigating some of the disruption generated by the ownership transition, although the estimates do not isolate bank networks from other differences between connected and unconnected firms. Taken together, the heterogeneity results point toward the importance of the institutional and political setting in which control was transferred: the largest profitability losses occurred in manufacturing, where conflict and resistance were strongest, while firms with prior state relationships and firms in strategic or resource-related activities followed more muted trajectories.

VI. Interpreting Heterogeneity and Potential Channels

Implementation, state capacity, and political conflict

Manufacturing was the principal arena of political and economic conflict surrounding the expansion of the social area. Textile firms were frequently transferred through interventions and requisitions initiated amid strikes, factory occupations, and disputes over production rather than through

[15]Note that these heterogeneity dimensions are not collinear with the sector fixed effects. Our sector classification follows the standard three-way division into primary, secondary, and tertiary activities, while the categories used here—manufacturing, strategic, and natural resource firms—are more fine-grained and cut across those broad sectors.

settled transfers of ownership and governance (Barros, 2013, pp. 169–260). In many industrial firms, the state assumed administration while compensation, legal ownership, and the division of authority between officials and workers remained unresolved. Owners, business associations, opposition parties, and courts contested these measures, while workers and parties within the UP pressed to retain intervened firms. Contemporary accounts accordingly describe manufacturing as the sector in which nationalization generated the most intense confrontation and contributed most directly to organized business resistance (de Vylder 1976, pp. 135–156; Silva 1996, pp. 40–57).

This process could affect performance through several overlapping channels. Uncertainty over ownership and governance could delay investment and operational decisions, labor-management conflict could interrupt production, and changes in managerial authority or personnel could weaken organizational continuity. Research on managers and management practices shows that managerial capabilities can have persistent effects on firm policies, productivity, and survival (Bertrand and Schoar, 2003; Giorcelli, 2019), while business-history research emphasizes the governance arrangements and networks supplying firms with monitoring, technical knowledge, finance, and other organizational services (Wilkins, 1988). Firms embedded in domestic business groups or dependent on imported inputs, foreign licenses, external finance, and established trading relationships could consequently be vulnerable when those arrangements were disrupted (Rojas, 2015). The baseline Allende-period estimates incorporate anticipation, negotiations, conflict, and administrative disruption as well as the period of state control itself. The larger decline among manufacturers may therefore reflect not only how firms operated after takeover but also the particularly conflictual process through which authority was contested and transferred. The annual reports identify board members but do not provide a consistently comparable record of senior managers or management practices, and this information has not been digitized across the panel. We consequently cannot separate managerial discontinuity from the other implementation channels described above.

Strategic and resource-related activities often entered the period with greater prior public involvement. Here, state capacity refers narrowly to sector-specific personnel, experience, and organizational routines for monitoring, financing, and coordinating production rather than to the general effectiveness of the Chilean state. CORFO and other public institutions had long participated in energy, mining, transport, telecommunications, and resource-based industries. State participation in copper under President Eduardo Frei provided experience with joint ownership and sectoral oversight (CORFO, 2009, pp. 105–106), while in fishing CORFO had already financed vessels and infrastructure, supplied technical assistance, and coordinated the restructuring of firms including Coloso and Indo (CORFO, 2009, pp. 126–133). Greater public control in these activities could therefore build on existing institutions and specialized knowledge, potentially reducing

organizational discontinuity. This interpretation remains necessarily cautious because manufacturing, strategic status, and natural-resource dependence combine differences in technology, market structure, capital intensity, worker organization, foreign exposure, political salience, and prior relations with the state. Nor were strategic and resource-sector transfers uniformly orderly: the copper nationalization generated an acute international dispute despite broad domestic support. The estimates do not isolate administrative capacity or political conflict from these other characteristics, but they indicate that outcomes varied with the institutional and political setting in which control was transformed, with the largest profitability losses occurring where implementation was especially contested and permanent governance remained unsettled.

International economic linkages

International economic relations provide another possible explanation for the decline in importing activity among targeted firms. After Allende took office, U.S. President Richard Nixon authorized covert actions intended to weaken the Chilean government as part of the wider Cold War effort to limit the spread of socialism in Latin America (Qureshi, 2009). The objective of making the economy "scream," revealed by National Security Decision Memorandum 93 and the Church Committee (U.S. Senate, 1975), included efforts to restrict bilateral assistance, multilateral lending, and private credit to Chile (Sigmund, 1974; Aldunate et al., 2024). At the same time, the UP's economic strategy explicitly sought to reduce foreign dependence and reorient production toward the domestic economy, which could itself alter international trade. Imports and exports are equilibrium outcomes reflecting decisions and constraints in both countries, so lower international activity cannot automatically be attributed to either U.S. pressure or Chilean policy alone. This interpretation connects to a broader historical literature showing that expropriation disputes were not confined to negotiations between foreign firms and host governments. Investors could mobilize their home governments, while aid, credit, trade policy, and geopolitical considerations became instruments in conflicts over property and compensation (Maurer, 2013, pp. 313–346).

We examine this broader context using bilateral trade flows from the International Monetary Fund's *Direction of Trade Statistics*. Trade is measured from the U.S. perspective: imports are U.S. imports from Chile reported on a cost, insurance, and freight (CIF) basis, whereas exports are U.S. exports to Chile reported on a free on board (FOB) basis. Following Abadie et al. (2010), we construct separate synthetic controls for the two flows using a donor pool of 18 other Latin American countries. For each outcome, the estimator reproduces the annual evolution of U.S.–Chile trade between 1955 and 1969 as closely as possible, without using observations from 1970

onward. The treatment period is 1970–1973, although 1970 combines months before and after Allende's inauguration and is best understood as a transition year.[16]

Figure 3 shows that U.S. imports from Chile and U.S. exports to Chile both diverged sharply from their synthetic counterparts after 1970. Imports from Chile declined immediately and remained substantially below the synthetic series throughout the UP period, whereas exports to Chile initially remained closer to the counterfactual but subsequently failed to follow its pronounced increase. Averaged over 1970–1973, observed imports from Chile were approximately 58 percent below their synthetic counterpart and observed exports to Chile approximately 43 percent below theirs. The earlier divergence in imports is consistent with escalating tensions surrounding the 1971 nationalization of the large copper companies (Edwards, 2025). The patterns remain when the synthetic controls use only odd-numbered pre-treatment years, reducing the risk of overfitting annual fluctuations (Ferman et al., 2020), and when trade is expressed in logarithms; Figure A5 in the Online Appendix reports these exercises.[17] Taken together, the evidence documents lower reported importing among targeted firms and a simultaneous decline in bilateral trade relative to Chile's earlier trajectory. The synthetic-control exercise remains descriptive and cannot separate U.S. policy, the domestic orientation of the UP program, macroeconomic disruption, changes in demand and supply, or a possible reallocation of U.S. trade toward donor countries. It provides context for the firm-level result without identifying the contribution of any single mechanism.

Objectives versus observed outcomes

The social area was not presented as a device for maximizing the accounting profits of firms. The UP sought to transfer strategic activities to public control, weaken domestic monopoly power and foreign economic dependence, coordinate investment, and redirect production toward basic goods consumed by lower-income households. As Barros (2013, pp. 26–45) emphasizes, the social area also had explicitly political and distributive objectives: it was intended to transfer economic power to workers, expand their participation in firm administration, and strengthen the constituency supporting the transition toward socialism. Accounting profitability was consequently only one dimension of a much broader project, and a decline in profits cannot by itself establish that the

[16]Appendix B reports the donor pool, weights, pre-treatment fit, and additional exercises. This analysis does not introduce a separate firm-level treatment comparison; it instead asks whether the decline in importing among targeted firms occurred within a broader deterioration of Chile's international economic relationships.

[17]Previous research also documents a decline in relationships between Chilean firms and U.S. banks after 1970 (Aldunate et al., 2024). This broader financial disengagement does not appear to be specific to firms targeted by the nationalization program: Table A12 shows no differential change in banking relationships between eventually nationalized and privately controlled firms.

government's distributive or political objectives failed.

The strategy nevertheless expected public control to improve production. Higher wages and public spending would activate idle capacity, while control over credit and investment would allow the state to accommodate the resulting demand and alter the composition of output. Worker participation was expected to strengthen incentives and prevent incumbent owners from withholding investment. This approach treated the main constraints as structural and political rather than as a shortage of private managerial expertise, but it also assumed that firms could be incorporated into a coordinated social area without prolonged uncertainty over ownership and access to essential inputs. That assumption proved particularly difficult to sustain in manufacturing, where transfers became entangled with labor conflict, owner resistance, litigation, and divisions within the governing coalition. Our firm-level evidence shows that targeted companies did not experience stronger short-run performance: return on assets declined substantially, with the clearest deterioration occurring in manufacturing. Estimates for sales over assets are also negative but considerably less precise and do not establish a general decline in physical production or capacity utilization. Profitability was not the government's sole objective, but it affected firms' ability to finance operations and investment and therefore remained relevant to the viability of the social area.

The importing result is similarly open to competing interpretations. Reducing dependence on foreign inputs and developing domestic substitutes were explicit elements of the UP strategy, so a lower probability of importing could represent a deliberate reorientation of production. The reports do not reveal whether an imported input was replaced domestically, became unnecessary because production changed, or became unavailable because firms lost access to foreign exchange, credit, suppliers, or technical relationships. Its coincidence with declining profitability is consistent with disruption to international business relationships, but it cannot distinguish voluntary substitution from externally or domestically imposed constraints. Many broader outcomes also remain outside the scope of the data. The annual reports do not measure employment security, worker participation in management, non-wage compensation, consumer access to basic goods, monopoly markups, or income distribution. Our evidence therefore does not provide a complete evaluation of the UP's project. It shows that the rapid expansion of public control was accompanied by substantial short-run profitability declines concentrated where implementation was most contested, while several intended distributive, political, and institutional outcomes cannot be evaluated with our data.

VII. Beyond the Firm

Wages in public firms

The decline in firm performance raises the question of how labor-market outcomes evolved in public firms. Because business reports do not report wages, we use the *Encuesta de Ocupación y Desocupación* (EOD), first mandated by the Central Bank in 1957 and administered by the University of Chile. The EOD is a probabilistic household survey of Santiago that collected labor-market information for household members aged 14 and older from 3,500 households during our period. It records the economic activity of the establishment and whether the worker's employer is a public administration, public firm, or private firm. Employment information was provided by the individual when present and otherwise could be reported by another household member.

We use the June waves from 1965 to 1973 and restrict the sample to employed individuals aged 18–65 whose employer is explicitly classified as either a public or a private firm, excluding public administrations and observations without a valid ownership classification. This leaves 32,276 worker-survey observations, including 1,514 public-firm and 30,762 private-firm workers. Although the survey covers Santiago only, the capital contained a large share of Chile's industrial employment and many of its largest enterprises, making it a relevant setting for examining public-private wage differentials. Econometrically, we estimate the following equation:

$$\log w_{it} = \sum_k \beta_k \left(P_{it} \times T_k\right) + \gamma x_i + \phi_t + \nu_{it}, \tag{3}$$

where w_{it} is the wage of worker i observed in year t, P_{it} indicates employment in a public firm, and T_k indicates each survey year between 1965 and 1973.[18] The vector x_i includes gender, age, age squared, and fixed effects for five education categories, while ϕ_t denotes survey-year fixed effects. The coefficients β_k describe composition-adjusted public-private wage differences within each year rather than following the same workers or identifying the effect of a particular firm's takeover. To account for changes in the industrial and occupational composition of public employment, we also replace survey-year fixed effects with economic-activity-by-year fixed effects, comparing public- and private-firm workers within the same activity and year. The Online Appendix additionally restricts the sample to activity-year cells containing both types of workers and includes broad-occupation-by-year fixed effects, and also examines whether observable characteristics predict

[18]Because the EOD consists of repeated cross sections and the comparison is between public- and private-firm workers within each survey year, all public-firm-by-year interactions can be estimated without an omitted year.

public-firm employment differently in 1965–1969 and 1972–1973 (Table A13). Sorting by age changes somewhat, but the relationship with education does not change systematically.

Figure 4 shows a substantial public-firm wage differential through 1971, followed by a sharp decline in 1972 and 1973. The average difference before 1972 is approximately 20 percent in the baseline specification and largely disappears in the final two waves. Table A14 shows that the differential declines by 17.4 log points in the baseline specification, 12.3 log points with economic-activity-by-year fixed effects, 11.6 log points in common-support activity-year cells, and 8.1 log points after additionally controlling for occupational composition. Clustering at the survey-segment level produces similar inference for the activity-adjusted estimate ($p = 0.003$). The same pattern appears for hourly wages, with estimated declines ranging from 18.2 to 9.9 log points across specifications, whereas hours worked do not decline differentially among public-firm workers. Changes in activity and occupational composition therefore account for part, but not all, of the deterioration in relative remuneration.

Figure A6 in the Online Appendix provides an additional diagnostic of the ownership classification. The share of workers classified as employed by public firms rises from approximately 4 percent before Allende's election to 6.2 percent in 1972, indicating that the survey responds to the expansion of public ownership. Its decline in 1973 and the absence of employer names, however, prevent us from determining whether workers in newly nationalized firms were reclassified immediately and consistently. More generally, the EOD cannot distinguish workers in always-public firms from those in newly nationalized firms. Together with the repeated cross-sectional design and conditioning on employment, this limitation means that the estimates describe changes in the composition-adjusted wage differential between workers classified as employed by public and private firms, rather than the causal wage effect of nationalization on the same workers.

Local exposure and electoral change

We next examine whether political support evolved differently across municipalities with different exposure to the nationalization program. Workers cannot be linked to particular plants or voting decisions, and plant locations were not randomly assigned, so the municipal analysis is a complementary descriptive exercise rather than a causal design. We construct a panel of 307 municipalities observed in four congressional elections—1961, 1965, 1969, and 1973—and three local elections—1963, 1967, and 1971. Following González and Prem (2025), we measure support for the governing coalition using the vote share received by Socialist and Communist candidates. The business reports identify the municipality of each plant owned by the 71 firms in our sample, allow-

ing us to construct three exposure measures: an indicator for hosting at least one plant belonging to an eventually nationalized firm, the number of such plants, and that number relative to municipal population in 1970. Econometrically, we estimate the following difference-in-differences model:

$$V_{mt} = \sum_k \beta_k (N_m \times T_k) + \phi_m + \phi_t + \epsilon_{mt}, \tag{4}$$

where V_{mt} denotes the UP vote share in municipality m and election t, N_m is one of the exposure measures, and T_k identifies election year. The 1969 congressional election is omitted. Municipality fixed effects absorb persistent local differences, election-year fixed effects absorb national changes, and standard errors are clustered by municipality. The coefficients therefore describe differential electoral trajectories across municipalities with different exposure levels.

Figure 5 shows that support for the UP evolved less favorably in more exposed municipalities by the 1973 congressional election. Municipalities hosting at least one nationalized plant experienced a relative decline of approximately 5 percentage points. The estimate for the 1971 local election is also negative but smaller and statistically insignificant, while estimates before 1969 are generally small and indistinguishable from zero; the pattern is similar when restricting the sample to congressional elections. The Online Appendix shows that the association remains negative under alternative panel specifications, province-by-election-year fixed effects, controls for local socioeconomic and institutional characteristics, and measures of exposure to other major UP policies. It also appears using synthetic difference-in-differences, while turnout does not decline differentially in more exposed municipalities. These exercises make an obvious pre-existing divergence and lower participation less plausible explanations, but they do not establish that exposure was exogenous. Municipalities hosting targeted plants may also have experienced distinctive strikes, shortages, political mobilization, or other consequences of the wider crisis, and the exposure measures cover only plants belonging to firms in our archival sample. We therefore interpret the results as a robust descriptive pattern: amid a nationwide crisis, support for the governing coalition evolved less favorably in municipalities with greater exposure to plants owned by nationalized firms.

VIII. Conclusion

Using new data, we show that firms targeted by Salvador Allende's nationalization program in Chile experienced a decline in profitability and reduced international engagement relative to firms that remained private. The deterioration in profitability was concentrated in manufacturing, where implementation was especially contested, whereas firms in strategic and natural-resource activities

followed more muted trajectories. We also find a decline in the public-sector wage differential and suggestive evidence of lower electoral support for the incumbent coalition in more exposed municipalities. These findings should be interpreted in light of the process through which public control was established. Nationalization unfolded amid labor conflict, unresolved governance, restrictions on credit and imported inputs, external pressure, and severe macroeconomic instability.

The evidence does not provide a complete welfare evaluation of the UP project. Profitability was not the government's sole objective, and the data do not measure several intended distributive, political, and developmental outcomes. Nor can the short period before the September 1973 coup reveal whether the documented changes persisted under the subsequent regime. The Chilean experience nevertheless shows that the consequences of large ownership transformations cannot be inferred from formal ownership alone. They depend on how control is transferred, the administrative capabilities already present, the intensity of political conflict, and firms' relationships with domestic and international partners.

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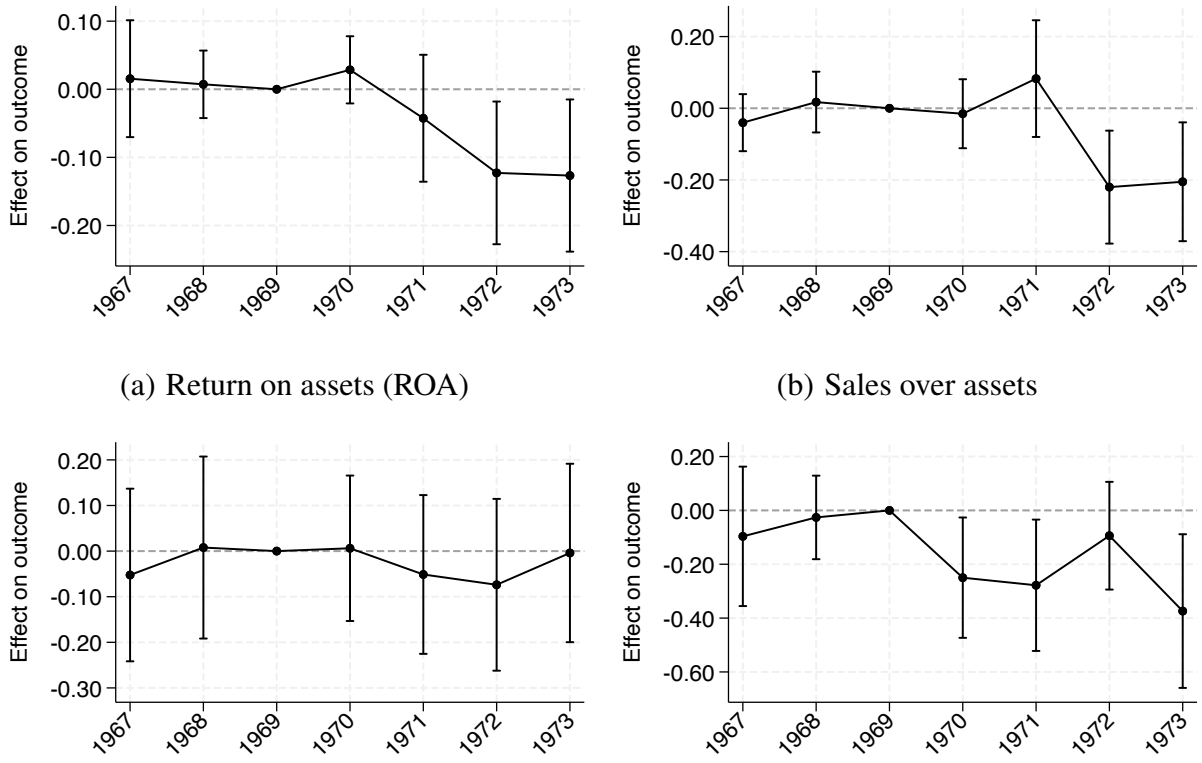
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Figure 1: Nationalization and firm performance



(a) Return on assets (ROA)

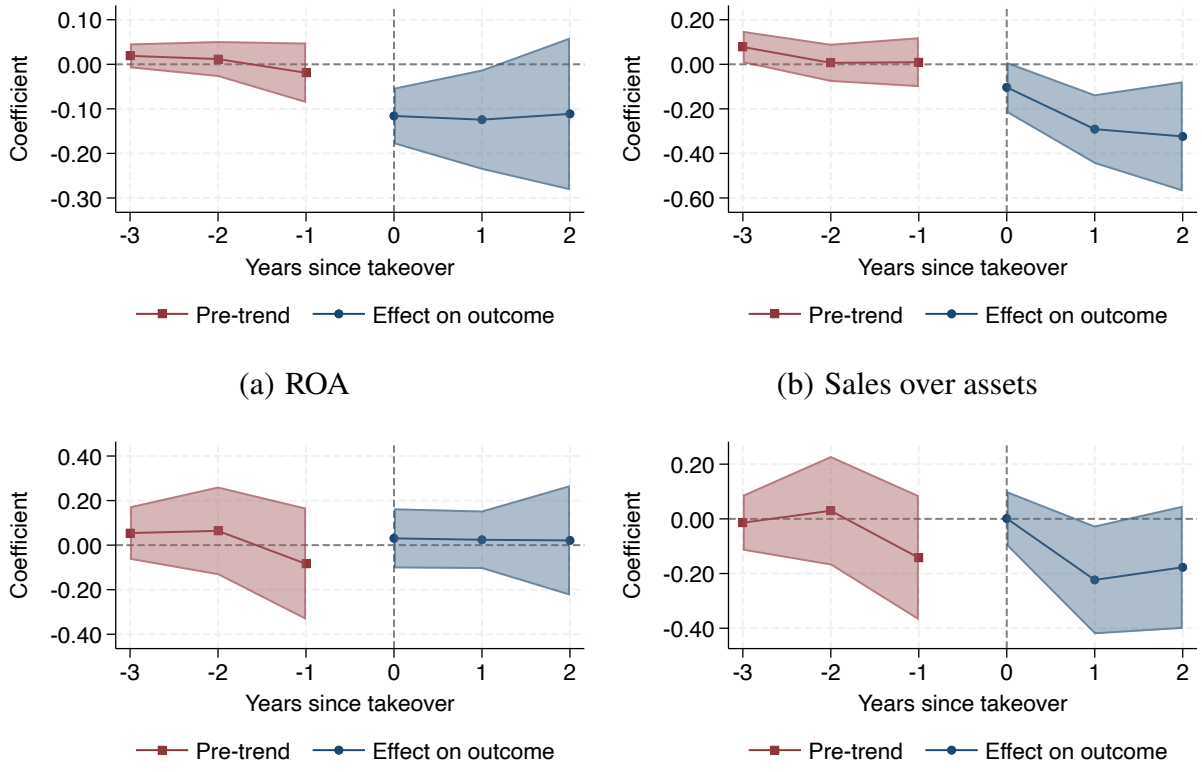
(b) Sales over assets

(c) Indicator exporting

(d) Indicator importing

Notes. Difference-in-differences estimates of the differential evolution of firms targeted by the UP nationalization program relative to firms that remained private. The sample consists of 71 firms observed annually between 1967 and 1973. The dependent variable is return on assets in panel A, sales over assets in panel B, an indicator equal to one if the firm exports in panel C, and an indicator equal to one if the firm imports in panel D. The omitted year is 1969, the last reporting year that unambiguously precedes Allende's election for all firms. The 1970 coefficient should be interpreted as a transition-year estimate because firms closed their fiscal years either in July or December. Black dots denote point estimates and vertical lines indicate 95 percent confidence intervals. All specifications include firm and sector-by-year fixed effects. Standard errors are clustered at the firm level.

Figure 2: Staggered difference-in-differences estimates



(a) ROA

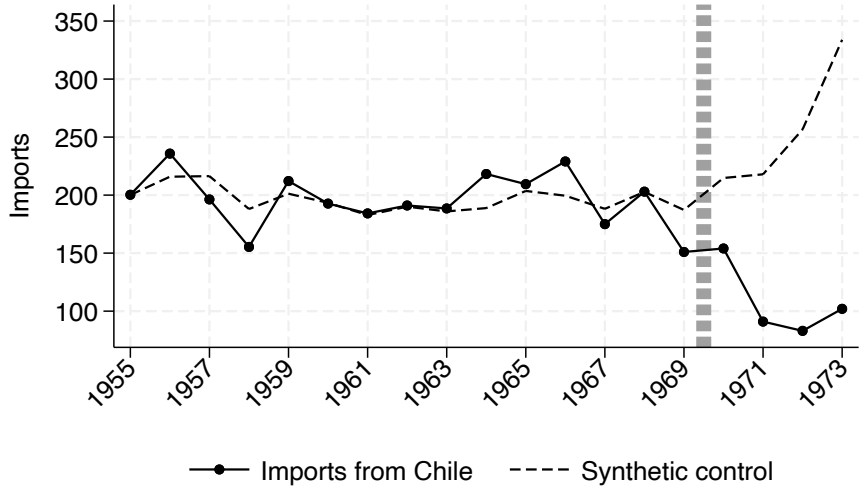
(b) Sales over assets

(c) Export

(d) Import

Notes. The figure reports event-study estimates using the imputation-based staggered difference-in-differences estimator proposed by Borusyak et al. (2024). The sample consists of 71 firms observed annually between 1967 and 1973. A firm becomes treated in the first fiscal-year report closing after its formal takeover date and remains treated thereafter. Event time zero therefore corresponds to the first report closing after takeover, negative values correspond to reports closing before takeover, and positive values correspond to subsequent reports. Because annual reports cover the preceding fiscal year, the event-time-zero observation may include months both before and after the transfer of control. The dependent variable is return on assets in panel A, sales over assets in panel B, an indicator equal to one if the firm exports in panel C, and an indicator equal to one if the firm imports in panel D. Squares and dots denote point estimates and shaded areas indicate 95 percent confidence intervals. All specifications include firm and sector-by-year fixed effects. Standard errors are clustered at the firm level.

Figure 3: Exploration of U.S.-Chile trade flows using a synthetic control method



(a) Imports from Chile



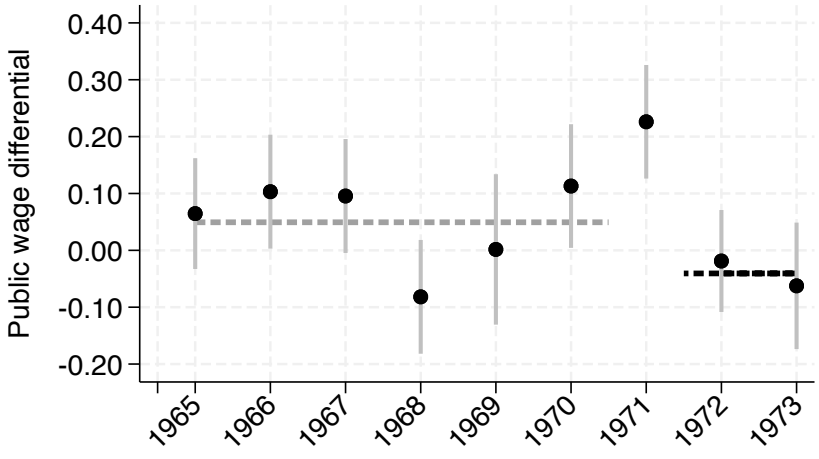
(b) Exports to Chile

Notes. Synthetic control estimates using a panel of Latin American countries and measures of exports (FOB) and imports (CIF) from the International Monetary Fund's Direction of Trade Statistics. The synthetic control weights are constructed using data from 1955 to 1969. The treatment period corresponds to 1970–1973. Trade flows are measured from the perspective of the United States, i.e. imports *from* Chile and exports *to* Chile.

Figure 4: Public-private wage differentials



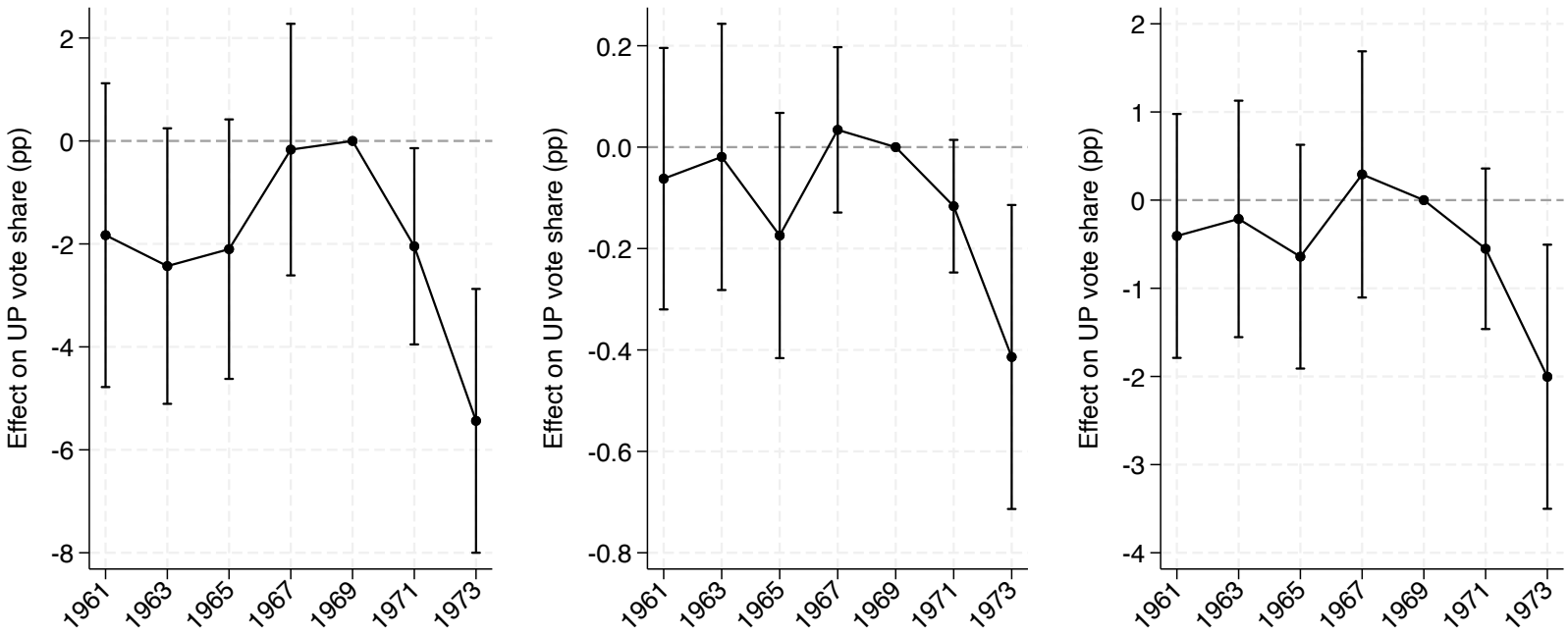
(a) Baseline specification



(b) Economic-activity-by-year fixed effects

Notes. The figure presents estimates of the public-private wage differential using repeated cross sections from the June waves of the *Encuesta de Ocupación y Desocupación* between 1965 and 1973. The sample includes employed individuals aged 18–65 whose workplace is explicitly classified by the survey as either a public or a private firm. The dependent variable is log wages. Both specifications control for gender, age, age squared, and education. Panel A includes survey-year fixed effects. Panel B includes economic-activity-by-year fixed effects, allowing wage changes to differ flexibly across economic activities. Activities classified as not well specified are excluded from panel B. Dots denote point estimates and vertical lines report 95 percent confidence intervals based on heteroskedasticity-robust standard errors.

Figure 5: Local exposure to nationalized plants and support for the incumbent UP coalition



(a) Indicator exposure to nationalization (b) Number of nationalized plants (c) Nationalized plants over 1970 pop.

Notes. These figures present difference-in-differences estimates of changes in electoral support for the left-wing coalition (y-axis) between 1961 and 1973. The definition of the cross-sectional treatment variable varies across panels: an indicator for municipalities hosting at least one plant belonging to a nationalized firm in panel A, the number of nationalized plants located in the municipality in panel B, and the number of nationalized plants per 1970 population in panel C. Black dots represent point estimates and vertical lines denote 95 percent confidence intervals. The dependent variable is the vote share of candidates affiliated with the Socialist and Communist parties in congressional elections (1961, 1965, 1969, and 1973) and local elections (1963, 1967, and 1971). All specifications include municipality and election-year fixed effects. Standard errors are clustered at the municipality level.

Table 1: Observed firms by year and nationalization status

	All firms	Non-nationalized	Nationalized	Early nationalization	Late nationalization
	(1)	(2)	(3)	(4)	(5)
1967	59	33	26	11	15
1968	62	34	28	12	16
1969	60	34	26	11	15
1970	59	34	25	11	14
1971	51	27	24	12	12
1972	49	25	24	12	12
1973	50	26	24	11	13
Distinct firms	71	39	32	14	18
Firm-year observations	390	213	177	80	97

Notes. The table reports the number of firms observed in each year of the estimation sample. Column 1 reports all firms, columns 2–3 divide the sample according to whether a firm was nationalized during the Allende administration, and columns 4–5 divide nationalized firms according to the timing of formal takeover. Early nationalizations correspond to firms formally taken over in 1971, while late nationalizations correspond to firms formally taken over after 1971. The last two rows report the number of distinct firms and firm-year observations in each group.

Table 2: Descriptive statistics in 1967-1969 by type of firm

	Nationalized	Non-Nationalized	Difference	Adding covariates	Year of nationalization
	(1)	(2)	(3)	(4)	(5)
Log total assets	17.88	15.82	2.06***		-0.12
	(0.91)	(2.04)	(0.39)		(0.27)
Log fixed assets	17.41	15.21	2.20***	0.18	-0.17
	(1.21)	(2.14)	(0.43)	(0.14)	(0.29)
Sales over assets	0.68	0.57	0.11	0.09	0.01
	(0.30)	(0.47)	(0.09)	(0.11)	(0.05)
ROA	0.18	0.18	0.00	-0.02	0.05
	(0.15)	(0.26)	(0.05)	(0.06)	(0.03)
Fixed over total assets	0.69	0.62	0.07	0.09	-0.04
	(0.22)	(0.28)	(0.06)	(0.07)	(0.05)
Log debt	16.48	14.39	2.09***	-0.53	0.60
	(2.83)	(2.76)	(0.66)	(0.67)	(0.83)
Leverage	0.39	0.39	-0.00	-0.05	-0.01
	(0.14)	(0.23)	(0.04)	(0.06)	(0.04)
Int'l market: Indicator exporter	0.43	0.23	0.20*	0.12	-0.08
	(0.50)	(0.42)	(0.11)	(0.14)	(0.14)
Int'l market: Indicator importer	0.81	0.46	0.36***	0.10	-0.03
	(0.39)	(0.50)	(0.10)	(0.11)	(0.06)
Sector: Primary	0.14	0.26	-0.12		-0.03
	(0.35)	(0.44)	(0.10)		(0.13)
Sector: Secondary	0.79	0.34	0.45***		0.00
	(0.41)	(0.47)	(0.11)		(0.12)
Sector: Tertiary	0.07	0.41	-0.33***		0.03
	(0.27)	(0.49)	(0.10)		(0.03)
Number of national banks	4.91	3.59	1.33**	-0.89	-0.90
	(2.01)	(2.69)	(0.62)	(0.78)	(0.56)
Number of international banks	3.04	1.33	1.72***	0.65	-0.32
	(1.65)	(1.47)	(0.39)	(0.40)	(0.53)
US banks	1.41	0.53	0.88***	0.57**	-0.25
	(1.01)	(0.71)	(0.22)	(0.25)	(0.26)
Firms	30	36	66	66	30

Notes. The table reports firm characteristics measured over 1967–1969 for the 66 firms with the information required for this exercise. Columns 1 and 2 report means and standard deviations for firms that were eventually nationalized and firms that remained privately owned, respectively. Column 3 reports the difference in means between nationalized and non-nationalized firms. Column 4 reports the same difference as in column 3 after adjusting for firm size (log total assets) and sector fixed effects; standard errors are reported in parentheses. Column 5 reports coefficients from a cross-sectional regression among the 30 firms that were eventually nationalized, where the dependent variable is the year of nationalization and the reported pre-Allende characteristics are included jointly; standard errors are reported in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 3: Difference-in-differences estimates

	Return on assets		Sales over assets		Indicator exporting		Indicator importing	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Ever nationalized × Allende period	-0.091***	-0.102***	-0.086*	-0.067	-0.070	-0.080	-0.196**	-0.239**
	(0.031)	(0.035)	(0.045)	(0.044)	(0.096)	(0.090)	(0.084)	(0.093)
Wild-bootstrap p-value	*0.01*	*0.01*	*0.05*	*0.13*	*0.49*	*0.41*	*0.01*	*0.01*
Observations	390	390	390	390	389	389	389	389
Firm fixed effects	Y	Y	Y	Y	Y	Y	Y	Y
Sector-year fixed effects	Y	Y	Y	Y	Y	Y	Y	Y
Covariates	N	Y	N	Y	N	Y	N	Y
Firms	71	71	71	71	71	71	71	71
Avg. dependent variable	0.180	0.180	0.616	0.616	0.330	0.330	0.612	0.612

Notes. The unit of observation is a firm-year. The table reports difference-in-differences estimates of the average Allende-period effect of being targeted by the UP nationalization program. The dependent variable is return on assets in columns 1–2, sales over assets in columns 3–4, an indicator equal to one if the firm exports in columns 5–6, and an indicator equal to one if the firm imports in columns 7–8. The treatment variable is the interaction between an indicator for firms that were nationalized during the Allende administration and a firm-specific Allende-period indicator. For firms reporting in December, the Allende-period indicator equals one beginning in 1970; for firms reporting in July, it equals one beginning in 1971. All specifications include firm and sector-by-year fixed effects. Even-numbered columns additionally control for firm size in the pre-treatment period, measured as average log total assets in 1967–1969, interacted with year fixed effects. The row labeled "Wild-bootstrap p-value" reports p-values obtained using a wild cluster bootstrap at the firm level with a small-sample correction. Conventional significance levels reported next to the coefficients are based on the firm-clustered standard errors shown in parentheses. Standard errors are clustered at the firm level. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 4: Nationalization and performance by type of firm

	Z = Manufacturing		Z = Strategic sector		Z = Nat. resources	
Dependent variable:	ROA	Importing	ROA	Importing	ROA	Importing
	(1)	(2)	(3)	(4)	(5)	(6)
Ever nationalized × Allende period	-0.033	-0.108	-0.153***	-0.083	-0.145***	-0.194**
	(0.038)	(0.104)	(0.041)	(0.077)	(0.033)	(0.085)
× Indicator for firm in Z	-0.130**	-0.153	0.172***	-0.367**	0.267***	-0.034
	(0.057)	(0.174)	(0.062)	(0.182)	(0.054)	(0.151)
Indicator for firm in Z × Allende period	0.089**	-0.023	-0.094**	0.274**	-0.024	0.302**
	(0.038)	(0.139)	(0.038)	(0.114)	(0.038)	(0.136)
Firm fixed effects	Y	Y	Y	Y	Y	Y
Sector-year fixed effects	Y	Y	Y	Y	Y	Y
Firms	71	71	71	71	71	71
Targeted firms in category Z	22	22	14	14	6	6
Avg. dependent variable	0.180	0.612	0.180	0.612	0.180	0.612

Notes. The unit of observation is a firm-year. The table reports difference-in-differences estimates with heterogeneous effects by type of firm: in manufacturing or not (columns 1–2), operating in strategic industries or not (columns 3–4), and operating in industries extracting natural resources or not (columns 5–6). The row labeled "× Indicator for firm in *Z*" reports the interaction between the baseline treatment effect (Ever nationalized × Allende period) and the indicator for firms in category Z. We use two measures of firm performance as dependent variables: return on assets (ROA) in odd columns and an indicator equal to one if the firm imports in even columns. Firms are classified as strategic following the original economic program of the Unidad Popular (Popular Unity, 1969). Standard errors are clustered at the firm level. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

ONLINE APPENDIX

Nationalization and Firm Performance in Allende's Chile

Felipe González Mounu Prem

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A Historical Addendum

A1 Nationalization in the UP program

The UP program divided the economy into social, mixed, and private areas and identified the activities intended for public control. de Vylder (1976, p. 112) describes the following as "the decisive passage in the UP program." We reproduce de Vylder's English translation of the original program (Popular Unity, 1969):

> "The process of transforming our economy will begin with a policy intended to create a dominant state area made up of enterprises already owned by the state plus those which are to be expropriated. As an initial step we will nationalize those basic riches—large-scale mining of copper, iron, nitrates, and others—which are now controlled by foreign capital and by domestic monopolies. This sector of nationalized activities will thus include the following activities:
>
> 1. large-scale mining of copper, nitrate, iodine, iron and coal;
> 2. the financial system of the country, in particular the private banks and insurance companies;
> 3. foreign trade;
> 4. large-scale enterprises and monopolies in the field of distribution;
> 5. strategic industrial monopolies;
> 6. in general those activities which condition the economic and social development of the country, such as the production and distribution of electrical energy, transportation by rail, air, and sea; communications; the production, refining and distribution of petroleum and its derivatives, including liquid gas; the iron and coal industry; cement, petrochemicals and heavy chemicals, cellulose, and paper.
>
> All these expropriations will always be carried out with full regard for the interests of the small shareholders."

A2 Legal and political complexity of nationalization

The UP initially sought to establish a statutory framework defining the social, mixed, and private areas of the economy. The government's proposal specified the sectors and large corporations eligible for transfer, preserved a private area for small and medium-sized firms, and established compensation provisions that treated small shareholders more favorably. Compensation would generally have been provided through long-term government bonds rather than immediate cash payments. The proposal did not obtain congressional approval, however, and the boundaries among the three areas remained the subject of continuing constitutional and political conflict (de Vylder, 1976, pp. 135–144).

In the absence of a comprehensive new law, the government relied on powers contained in existing legislation. A decree-law dating from 1932 permitted the executive to act against firms

producing goods considered essential when production was interrupted, capacity remained unused, or price regulations were violated. The available measures included expropriation, requisition, and intervention. Expropriation transferred ownership and required compensation, whereas requisition and intervention transferred administrative control without resolving the firm's permanent ownership. Because expropriation required substantial immediate resources, the government more frequently relied on the latter instruments, which were formally temporary but could remain in place for extended periods.

These administrative measures operated alongside negotiated purchases, public acquisitions of shares, and proposals for mixed ownership. The route followed therefore varied across firms and could change over time: intervention might precede negotiations, a share-purchase campaign, or a proposed mixed enterprise. The resulting social area was not produced by a single legal mechanism, but by the interaction of government policy with congressional opposition, judicial challenges, owners' willingness to negotiate, worker mobilization, and foreign involvement (de Vylder, 1976, pp. 135–144).

B Synthetic Control Analysis of U.S.-Chile Trade

We use annual bilateral trade data from the International Monetary Fund's *Direction of Trade Statistics* to compare U.S.–Chile trade with a synthetic counterfactual constructed from other Latin American countries. Trade flows are measured from the perspective of the United States: imports are U.S. imports from Chile reported on a CIF basis, while exports are U.S. exports to Chile reported on an FOB basis. Chile is the treated unit. After excluding Cuba, the donor pool contains 18 Latin American countries.[19]

We estimate separate synthetic controls for imports and exports. For each outcome, the synthetic control is a weighted average of the countries in the donor pool:

$$Y_t^{SC} = \sum_{j=1}^{J} w_j Y_{jt}, \tag{5}$$

where Y_{jt} is the corresponding U.S. trade flow with donor country j in year t. The country weights w_j are constrained to be nonnegative and to sum to one. The estimator selects these weights to minimize the discrepancy between Chile and its synthetic counterpart during the pre-treatment period. We use the outcome in every year from 1955 through 1969 as a predictor, and estimate the weights separately for imports and exports. No observations from 1970 onward are used to construct the synthetic control.

The treatment period begins in 1970, the year of Allende's election and inauguration, and extends through 1973. Because the trade data are annual, the 1970 observation combines months before and after the change in administration and should be interpreted as a transition-year outcome. For each post-1969 year, the estimated gap is the observed trade flow between the United

[19]The donor pool comprises Argentina, Bolivia, Brazil, Colombia, Costa Rica, the Dominican Republic, Ecuador, El Salvador, Guatemala, Haiti, Honduras, Mexico, Nicaragua, Panama, Paraguay, Peru, Uruguay, and Venezuela.

States and Chile minus the trade flow predicted by the synthetic control.

Table A15 reports the estimated donor weights and measures of pre-treatment fit. The synthetic control for U.S. imports from Chile places positive weight on Uruguay, Ecuador, Colombia, Brazil, and Venezuela. The export counterfactual places most of its weight on Honduras, with additional contributions from Argentina, Ecuador, Mexico, and Brazil. All remaining donor countries receive zero weight. The pre-treatment root mean squared prediction error is 18.7 for imports and 22.2 for exports, corresponding to 9.5 and 11.1 percent of their respective pre-treatment means. Thus, although the synthetic series does not reproduce every annual fluctuation, it tracks the broad level and evolution of Chilean trade before 1970.

The divergence becomes considerably larger after 1970. During 1970–1973, observed U.S. imports from Chile average 107.5, compared with 255.8 for the synthetic control, producing an average gap of −148.3. Observed U.S. exports to Chile average 239.8, compared with a synthetic average of 423.8, producing a gap of −184.0. Relative to the corresponding synthetic averages, actual imports and exports are approximately 58 and 43 percent lower. The post-treatment RMSPE is more than eight times its pre-treatment value for both outcomes.

Figure A5 examines the sensitivity of these patterns to alternative specifications. To reduce concerns that using every pre-treatment year may overfit short-run fluctuations, we construct alternative synthetic controls using only odd-numbered pre-treatment years. We also estimate the models using logarithmic trade flows. The divergence between Chile and its synthetic counterpart after 1970 remains visible across these alternatives.

The exercise should nevertheless be interpreted as descriptive rather than as an isolated causal estimate of the Allende administration's effect on trade. Bilateral trade reflects policies and economic conditions in both countries, and the donor countries may themselves have experienced changes in their trading relationships with the United States. Moreover, we do not use in-space placebo estimates to attach conventional statistical significance to the post-1970 gaps. The synthetic-control evidence therefore documents a sharp divergence in U.S.–Chile trade relative to a transparent data-driven counterfactual and provides suggestive evidence about the deterioration of international economic relations during the UP period.

C Additional Evidence on Electoral Change

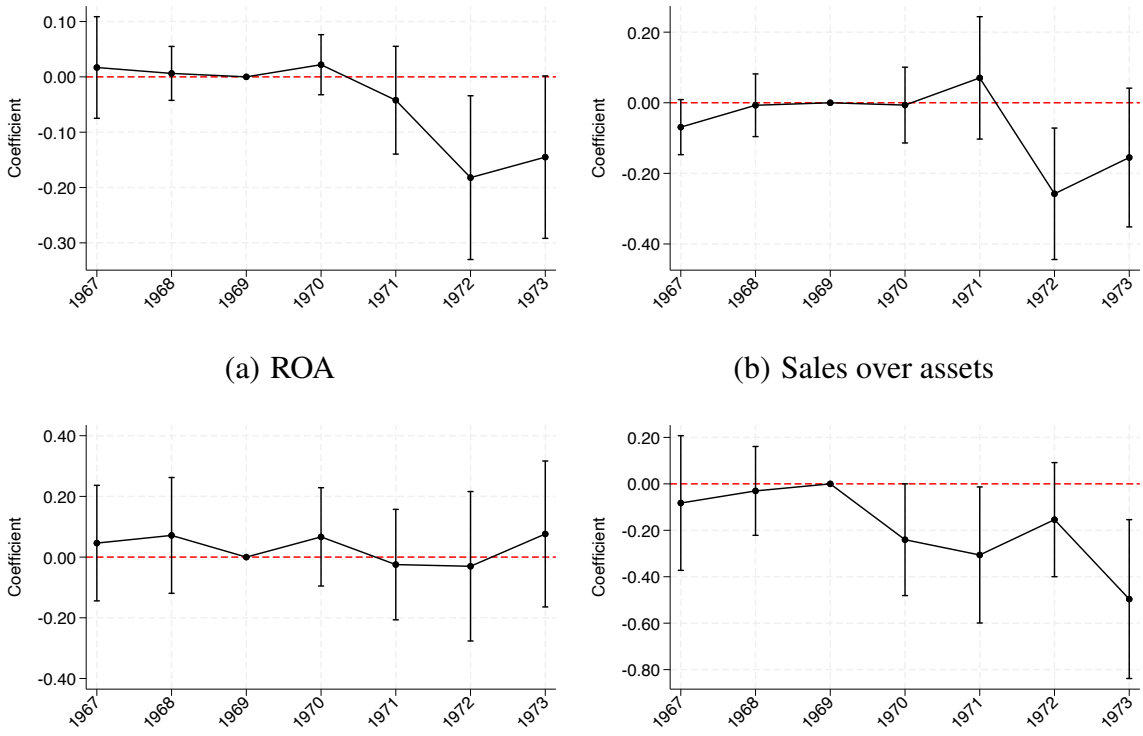
This section presents additional evidence on the relationship between local exposure to plants owned by nationalized firms and the evolution of electoral support for the UP coalition. As emphasized, the location of these plants was not randomly assigned, and the available data do not allow us to link individual workers, firms, and voting decisions. We therefore interpret the results as documenting differential electoral patterns rather than isolating the causal effect of nationalization on voting. Figure A7 first restricts the flexible specification to the four congressional elections held in 1961, 1965, 1969, and 1973. Municipalities with greater exposure experienced a less favorable evolution of UP support between 1969 and 1973 under all three measures of exposure, while the coefficients for earlier congressional elections are generally smaller and statistically indistinguishable from zero. The similarity with the results combining congressional and local elections indicates that the main pattern is not driven by differences across types of electoral contests.

Table A16 reports a parsimonious specification in which the differential change is restricted to the 1973 election. The columns use, respectively, an indicator for hosting at least one plant owned by a nationalized firm, the number of such plants, and the number of plants relative to municipal population. Greater exposure is associated with a relative decline in UP support of 1–4 percentage points. Table A17 examines the stability of this relationship across alternative samples and specifications. The estimates remain negative when restricting the sample to congressional elections, collapsing the panel into periods before and during the UP government, and allowing for flexible differential movements before 1970. They are also similar after controlling for local education and labor-market characteristics, institutional and state presence, spatial dependence across municipalities, and the presence of plants owned by large firms that remained private. The pattern survives the inclusion of province-by-election-year fixed effects and the exclusion of provincial capitals.

Because the UP government implemented several major policies at the same time, Table A18 examines whether the electoral pattern is accounted for by other contemporaneous interventions or structural changes. The specifications control for the intensity of agrarian reform (Cuesta et al., 2017; González and Vial, 2021), local exposure to trade protection (Lederman, 2005; Cuesta et al., 2015), and exposure to the government's milk-distribution program (Frens-String, 2021; González and Prem, 2025). Additional exercises account for the local importance of mining and agriculture, distance to university campuses, the municipal illiteracy rate in 1970, and migration between 1960 and 1970 (Cousiño, 2001). The estimated association between exposure to plants owned by nationalized firms and electoral change remains negative across these specifications. These results indicate that the pattern is not mechanically explained by the measured exposure to other major policies or by several dimensions of local economic structure, although the available controls cannot capture every local consequence of the broader economic and political crisis.

Finally, Figure A8 applies the synthetic difference-in-differences estimator of Arkhangelsky et al. (2021), comparing municipalities hosting at least one nationalized plant with a weighted combination of municipalities chosen to reproduce their earlier electoral trajectories. Panel A shows a similar relative decline in UP support. Panel B finds no systematic relative decline in voter turnout in more exposed municipalities, suggesting that lower participation does not account for the vote-share pattern. Taken together, the exercises show a stable association between local exposure and a less favorable evolution of UP support by 1973. Nevertheless, plant locations reflect Chile's pre-existing industrial geography, the exposure measures cover only firms represented in our archival sample, and exposed municipalities may have experienced other distinctive shocks. We therefore interpret the evidence as a robust descriptive relationship rather than a causal estimate of the political consequences of nationalization.

Figure A1: Difference-in-differences results, controlling for the size of firms



(a) ROA

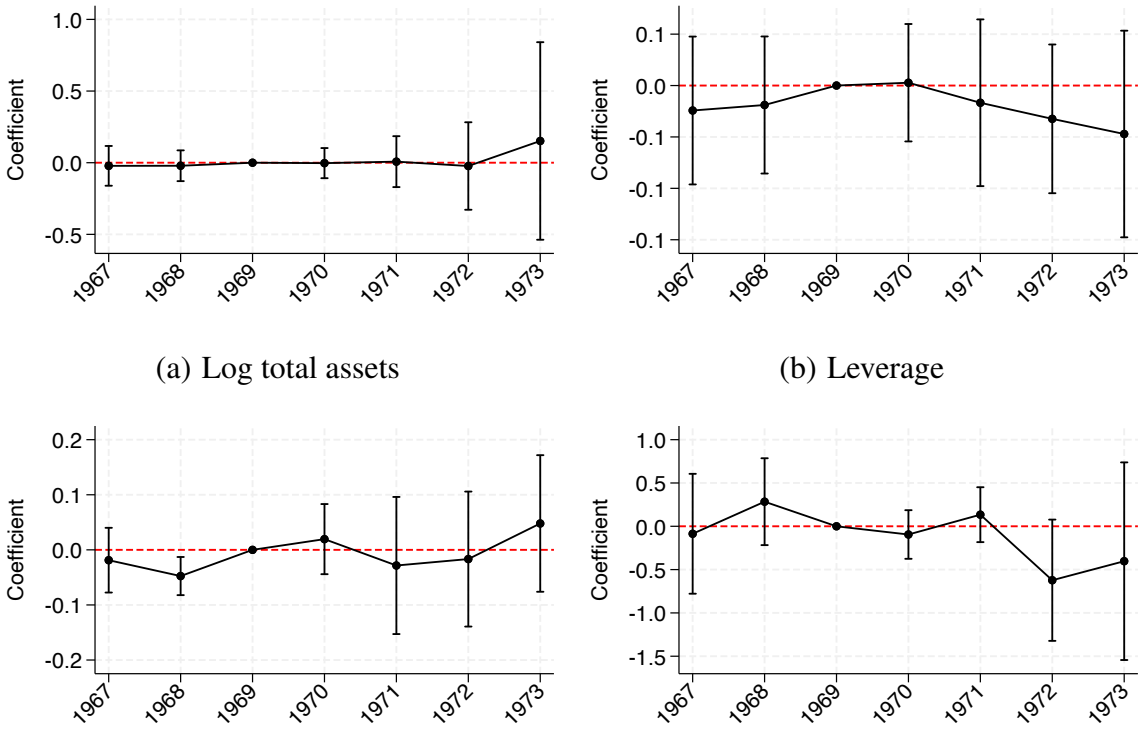
(b) Sales over assets

(c) Export

(d) Import

Notes. Difference-in-differences estimates of the differential evolution of firms targeted by the UP nationalization program relative to firms that remained private. The sample consists of 71 firms observed annually between 1967 and 1973. The dependent variable is return on assets in panel A, sales over assets in panel B, an indicator equal to one if the firm exports in panel C, and an indicator equal to one if the firm imports in panel D. The omitted year is 1969, the last reporting year that unambiguously precedes Allende's election for all firms. The 1970 coefficient should be interpreted as a transition-year estimate because firms closed their fiscal years either around June or July or in December. Black dots denote point estimates and vertical lines indicate 95 percent confidence intervals. All specifications include firm and sector-by-year fixed effects and additionally control for baseline firm size, measured as average log total assets in 1967–1969, interacted with year fixed effects. Standard errors are clustered at the firm level.

Figure A2: Difference-in-differences estimates, additional outcomes



(a) Log total assets

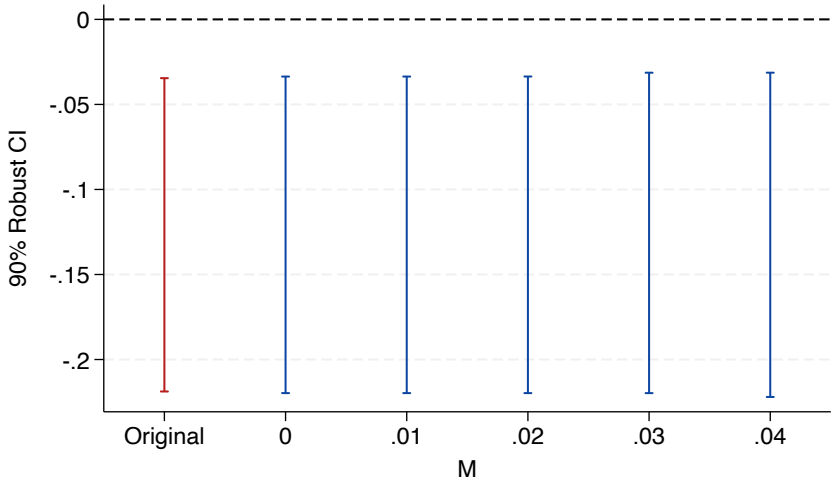
(b) Leverage

(c) Fixed assets over total assets

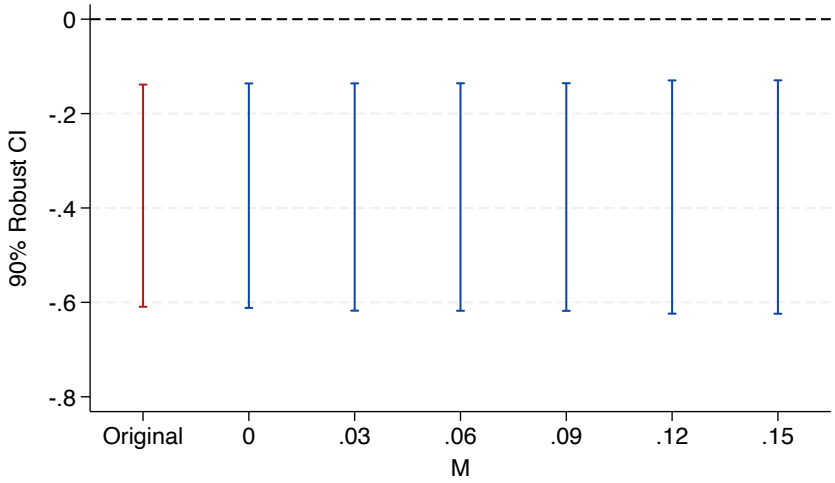
(d) Log sales

Notes. Difference-in-differences estimates of the differential evolution of firms ultimately targeted by the UP nationalization program relative to firms that remained private. The nationalization program began after Salvador Allende took office in November 1970. The dependent variable is log total assets in panel A, leverage (debt over assets) in panel B, fixed assets over total assets in panel C, and log sales in panel D. Black dots represent point estimates and vertical lines denote 95 percent confidence intervals. All specifications include firm and sector-by-year fixed effects. Standard errors are clustered at the firm level.

Figure A3: Honest difference-in-differences



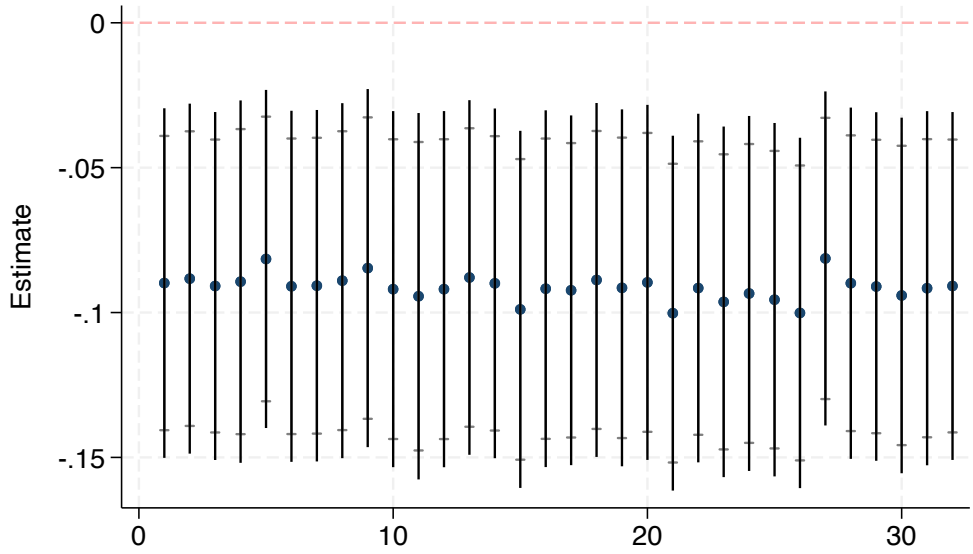
(a) Return on assets



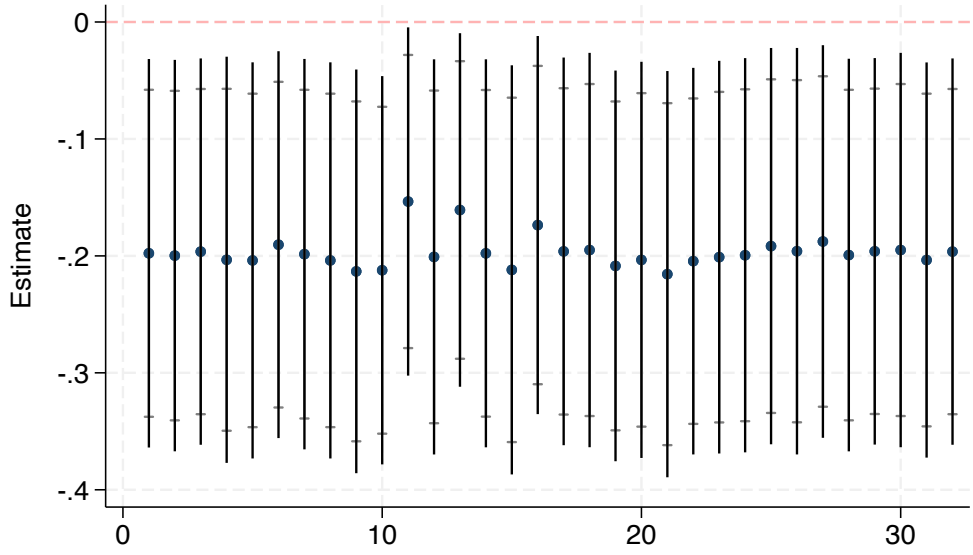
(b) Indicator importing

Notes. This figure assesses the robustness of our main difference-in-differences estimates to potential violations of the parallel trends assumption using the honest difference-in-differences approach of Rambachan and Roth (2023). Panel A reports results for return on assets (ROA), and panel B for an indicator equal to one if the firm imports. The y-axis shows 90% robust confidence intervals, while the x-axis reports the original point estimate together with estimates under different values of the hyperparameter M which bounds post-treatment deviations from parallel trends. The labels in the x-axis, i.e. 0-0.04 in panel A and 0-0.15 in panel B, are values of M. The vertical lines provide the honest-DiD 10% confidence interval for each M. The "original" estimates in both panels correspond to a standard 10% confidence interval in the specifications presented in columns 2 and 8 in Table 3. All specifications include firm and sector-by-year fixed effects, and standard errors are clustered at the firm level.

Figure A4: Excluding single firms



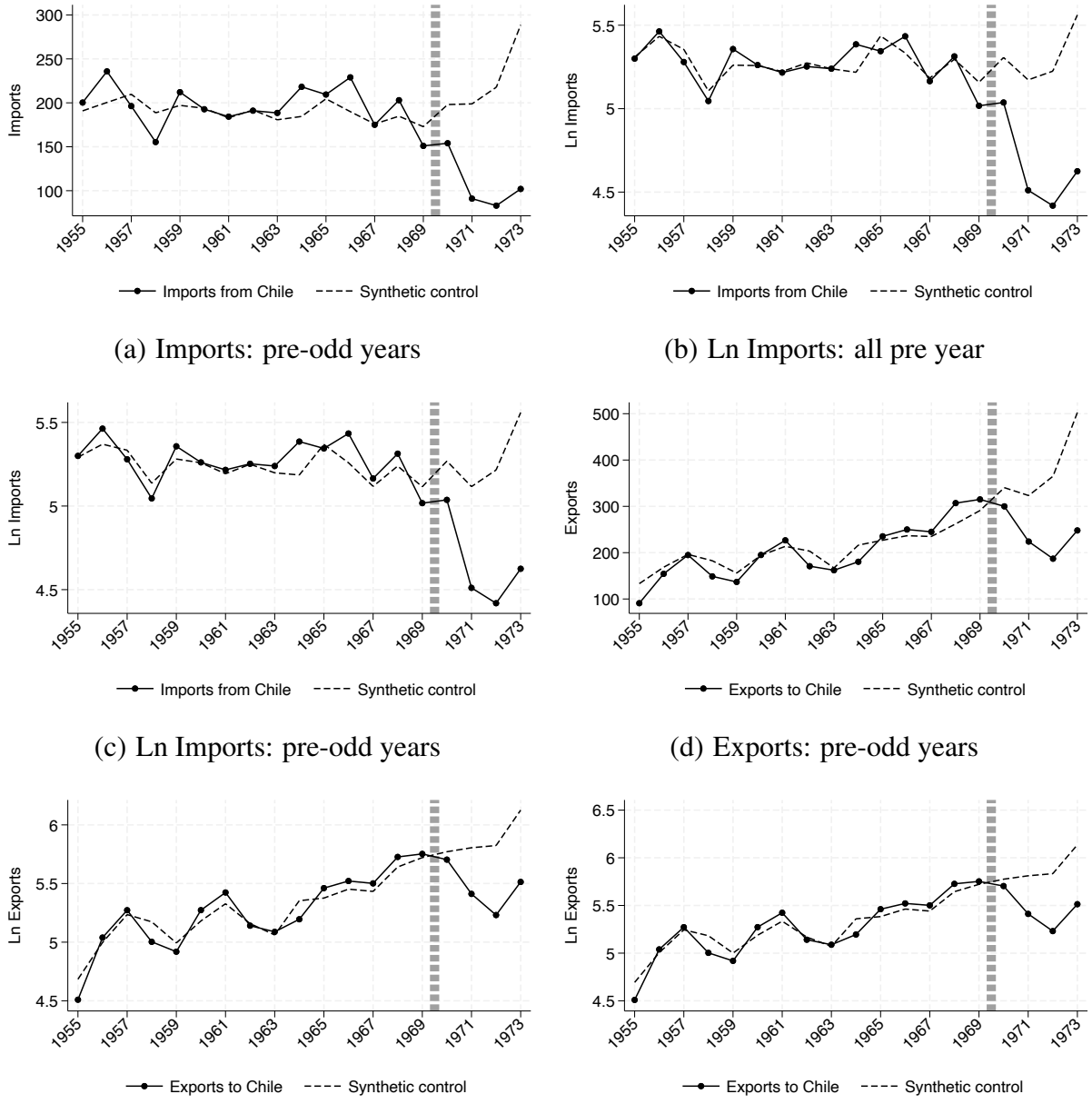
(a) ROA



(b) Import

Notes. Leave-one-out difference-in-differences estimates for the effect of nationalization on firm performance. Panel A reports estimates using return on assets (ROA) as the dependent variable, and panel B uses an indicator equal to one if the firm imports. In each estimation, one of the 32 nationalized firms is excluded from the sample; the x-axis identifies the excluded firm, and the y-axis reports the corresponding estimate. All specifications include firm and sector-by-year fixed effects. Standard errors are clustered at the firm level.

Figure A5: Additional synthetic control results



(a) Imports: pre-odd years

(b) Ln Imports: all pre year

(c) Ln Imports: pre-odd years

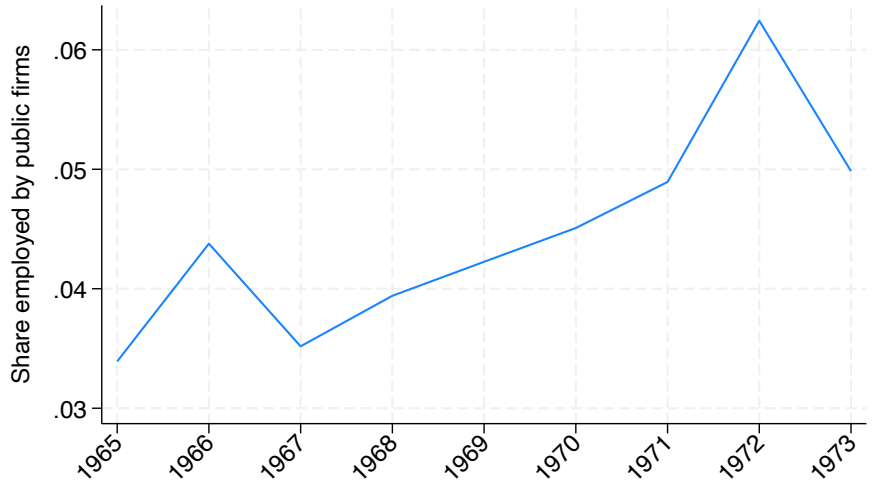
(d) Exports: pre-odd years

(e) Ln Exports: all pre year

(f) Ln Exports: pre-odd years

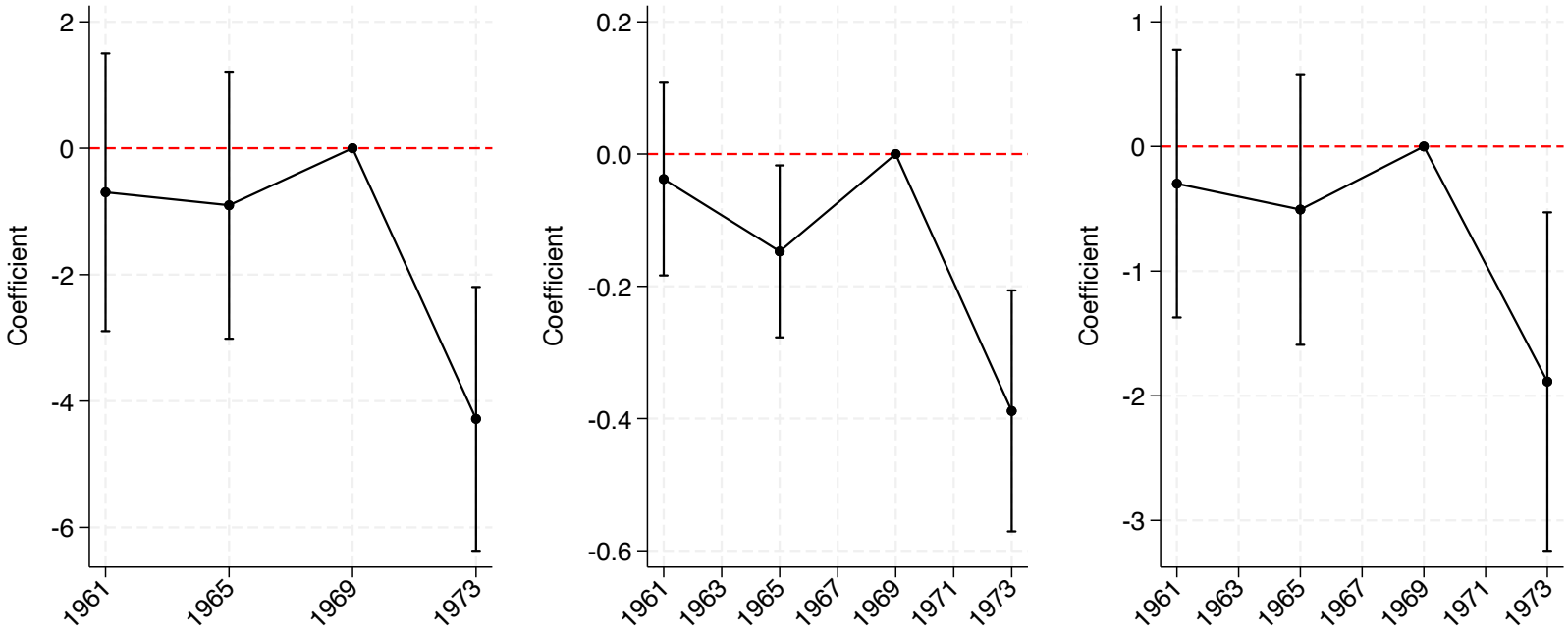
Notes. Synthetic control estimates using a panel of all Latin American countries and trade data from the International Monetary Fund (Direction of Trade Statistics). Panels A, C, D, and F construct the synthetic control using only odd years in the pre-treatment period. Panels B and C use log imports (CIF) as the dependent variable, while panels E and F use log exports (FOB); remaining panels use trade levels as in Figure 3. The pre-treatment period spans 1955-1969, and the treatment period is 1970–1973.

Figure A6: Share of workers classified as employed by public firms



Notes. The figure reports the annual share of workers classified by the *Encuesta de Ocupación y Desocupación* as employed by public firms. We use the June survey waves from 1965 to 1973 and restrict the sample to employed individuals aged 18–65 whose workplace is explicitly classified as either a public or a private firm. The data are repeated cross sections rather than a panel of workers. The increase in the reported public-firm employment share during the UP period indicates that the survey classification responds to the expansion of the public sector, but it does not establish that workers in newly nationalized firms were reclassified immediately or without measurement error.

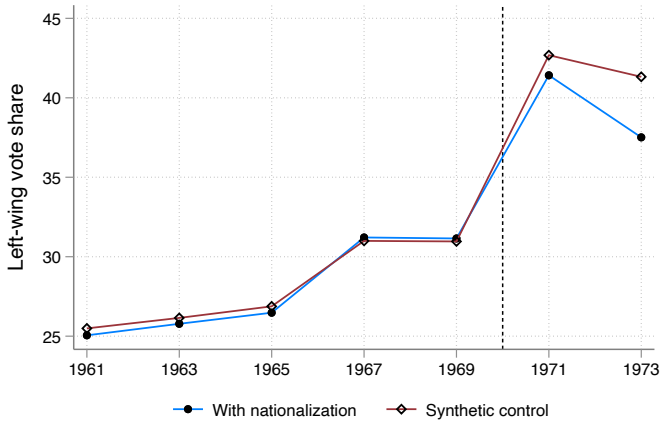
Figure A7: Nationalizations and UP voting, only Congress



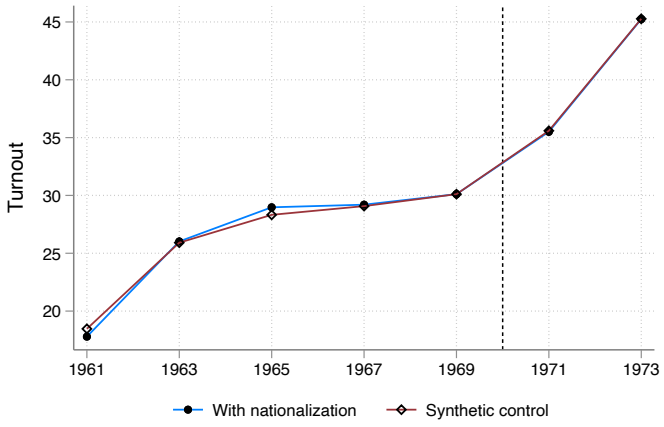
(a) Indicator exposure to nationalization (b) Number of nationalized plants (c) Nationalized plants over 1970 pop.

Notes. This figure presents difference-in-differences estimates of changes in electoral support for the left-wing coalition using congressional elections only. The sample includes 307 municipalities observed in four congressional elections (1961, 1965, 1969, and 1973). The definition of the cross-sectional treatment variable varies across panels: an indicator for municipalities hosting at least one plant belonging to a nationalized firm in panel A, the number of nationalized plants located in the municipality in panel B, and the number of nationalized plants per 1970 population in panel C. Black dots represent point estimates and vertical lines denote 95 percent confidence intervals. The dependent variable is the vote share of candidates affiliated with the Socialist and Communist parties. All specifications include municipality and election-year fixed effects. Standard errors are clustered at the municipality level.

Figure A8: Electoral change using synthetic difference-in-differences



(a) Left-wing vote share



(b) Turnout

Notes. These figures present synthetic difference-in-differences estimates of changes in electoral support for the left-wing coalition (y-axis, panel A) and turnout (panel B) between 1961 and 1973 (Arkhangelsky et al., 2021). The "With nationalization" and "Without nationalization" groups represent group averages of treated municpalities versus their synthetic controls constructed from donor municipalities. The definition of the cross-sectional treatment variable is an indicator for municipalities hosting at least one plant belonging to a nationalized firm. The dependent variable is the vote share of candidates affiliated with the Socialist and Communist parties (panel A) or turnout (panel B) in congressional elections (1961, 1965, 1969, and 1973) and local elections (1963, 1967, and 1971). Point estimates are -3.7 in panel A with a 95 percent confidence interval of $[-6.1, -1.3]$ and -0.10 in panel B with a 95 percent confidence interval of $[-2.2, 2.0]$.

Table A1: Selection into the firm-level estimation sample

	Estimation sample (1)	Excluded from estimation sample (2)	Difference (3)
Nationalization status			
Ever nationalized	0.45	0.38	0.08
	(0.50)	(0.50)	(0.14)
Size, performance, and balance-sheet characteristics			
Log total assets	16.75	15.95	0.80
	(1.86)	(4.52)	(1.13)
Log fixed assets	16.17	15.41	0.76
	(2.05)	(4.47)	(1.12)
Sales over assets	0.65	0.70	-0.04
	(0.40)	(0.65)	(0.17)
ROA	0.20	0.10	0.10*
	(0.23)	(0.11)	(0.06)
Fixed over total assets	0.64	0.58	0.06
	(0.25)	(0.29)	(0.08)
Log debt	15.41	15.06	0.35
	(2.47)	(4.52)	(1.21)
Leverage	0.41	0.34	0.07
	(0.19)	(0.20)	(0.06)
International-market participation			
Exporter	0.42	0.19	0.24**
	(0.50)	(0.40)	(0.12)
Importer	0.70	0.69	0.02
	(0.46)	(0.48)	(0.13)
Sector			
Primary	0.20	0.19	0.01
	(0.40)	(0.40)	(0.11)
Secondary	0.55	0.56	-0.01
	(0.50)	(0.51)	(0.14)
Tertiary	0.25	0.25	0.00
	(0.44)	(0.45)	(0.12)
Geographic presence			
Santiago Metropolitan Region	0.76	0.62	0.14
	(0.43)	(0.50)	(0.13)
Northern Chile	0.25	0.12	0.13
	(0.44)	(0.34)	(0.10)
Southern Chile	0.34	0.50	-0.16
	(0.48)	(0.52)	(0.14)
Banking relationships			
Number of national banks	4.16	5.08	-0.92
	(2.28)	(2.52)	(0.67)
Number of international banks	2.04	1.94	0.10
	(1.66)	(1.78)	(0.48)
Number of U.S. banks	0.87	0.92	-0.05
	(0.92)	(0.95)	(0.26)
Firms	71	16	
Firms included in joint test			74
Joint significance, p-value			0.973

Notes. This table compares firms entering the baseline estimation sample with firms in the archival dataset that do not enter the estimation sample. The unit of observation is a firm. Continuous firm characteristics are averages constructed from available reports between 1967 and 1970. Exporter and importer are indicators equal to one if the firm reports the corresponding activity in at least one available report during this period. The geographic indicators equal one if the firm has at least one recorded operation in the Santiago Metropolitan Region, northern Chile, or southern Chile. Because firms may operate in multiple locations, the geographic indicators are not mutually exclusive. Columns 1 and 2 report means, with cross-firm standard deviations in parentheses. Column 3 reports the difference between firms in the estimation sample and excluded firms, with heteroskedasticity-robust standard errors in parentheses. The joint-test row reports the p-value from a linear probability model of sample inclusion on eventual nationalization, log total assets, ROA, leverage, exporter and importer status, sector indicators, and geographic presence. The joint test uses the 74 firms with complete information for all included covariates. The number of firms underlying individual rows may vary because some characteristics are unavailable for firms without reports during 1967–1970. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A2: Analysis of missing business reports

	Dependent variable: Number of missing years					
	All years		Pre-Allende years		Allende years	
	(1)	(2)	(3)	(4)	(5)	(6)
Ever nationalized	-0.07	-0.11	0.18	0.20	-0.25	-0.31
	(0.39)	(0.52)	(0.23)	(0.28)	(0.27)	(0.37)
Log total assets		0.14		0.05		0.09
		(0.14)		(0.08)		(0.12)
Indicator for importing		-0.32		-0.07		-0.24
		(0.66)		(0.38)		(0.48)
Number of U.S. banks		-0.59		-0.39		-0.20
		(0.58)		(0.32)		(0.39)
Return on assets		1.03		0.40		0.64
		(0.87)		(0.57)		(0.61)
Sales over assets		0.17		0.25		-0.08
		(0.45)		(0.34)		(0.45)
Firms	71	71	71	71	71	71
Avg. dependent variable	1.51	1.51	0.62	0.62	0.89	0.89
Joint test p-value	–	0.61	–	0.48	–	0.68

Notes. The unit of observation is a firm. The dependent variable is the number of years with missing firm-level information: total missing years (columns 1–2), missing years in the pre-Allende period (columns 3–4), and missing years in the Allende period (columns 5–6). Regressions include nationalization status and, in even columns, baseline firm characteristics measured before 1970. Robust standard errors are reported in parentheses. The last row reports p-values from a joint test that all coefficients are equal to zero. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A3: Pre-Allende predictors of selection into nationalization

	Dependent variable: Nationalized firm				
	Firm size	Firm performance	International exposure	Balance sheet and banks	Joint model
	(1)	(2)	(3)	(4)	(5)
Log total assets	0.115***	0.110***	0.104**	0.104**	0.119**
	(0.040)	(0.039)	(0.050)	(0.050)	(0.055)
ROA		-0.264			-0.201
		(0.220)			(0.313)
Sales over assets		0.144			0.256
		(0.142)			(0.172)
Share of pre-Allende years exporting			0.048		0.055
			(0.152)		(0.133)
Share of pre-Allende years importing			0.056		-0.011
			(0.147)		(0.183)
Fixed assets over total assets				0.277	0.321
				(0.210)	(0.240)
Leverage				-0.224	-0.407
				(0.274)	(0.320)
Number of national banks				-0.080**	-0.085**
				(0.036)	(0.036)
Number of international banks				0.046	0.038
				(0.073)	(0.078)
Number of U.S. banks				0.098	0.091
				(0.103)	(0.112)
Firms	67	66	67	63	63
R-squared	0.320	0.336	0.323	0.436	0.456
Avg. dependent variable	0.463	0.455	0.463	0.476	0.476
Sector fixed effects	Y	Y	Y	Y	Y
p-value: log total assets	0.005				
p-value: additional covariates		0.420	0.878	0.037	0.104

Notes. The unit of observation is a firm. The dependent variable is an indicator equal to one if the firm was eventually nationalized under the UP program. Explanatory variables are firm-level averages calculated using available observations from 1967–1969. The exporting and importing variables measure the share of observed pre-Allende years in which the firm reported each activity. All columns report linear probability models and include sector fixed effects. Column 1 includes pre-Allende firm size. Column 2 adds measures of firm performance, column 3 adds measures of international exposure, and column 4 adds measures of balance-sheet composition and banking relationships. Column 5 includes all reported covariates jointly. The row labeled "p-value: additional covariates" reports a joint test of the covariates added to log total assets and the sector fixed effects in each specification. The sample varies across columns because some pre-Allende characteristics are unavailable for a small number of firms. Heteroskedasticity-robust standard errors are reported in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A4: Differential pre-trends

	ROA	Sales over assets	Export	Import
	(1)	(2)	(3)	(4)
Ever nationalized × Linear trend	0.005	0.031	0.047	0.032
	(0.021)	(0.020)	(0.047)	(0.065)
Observations	179	179	178	178
Firm fixed effects	Y	Y	Y	Y
Sector-year fixed effects	Y	Y	Y	Y
Firms	64	64	64	64
Avg. dependent variable	0.181	0.615	0.315	0.612

Notes. The unit of observation is a firm-year. The table reports difference-in-differences estimates testing for differential pre-trends prior to the Allende government. The sample is restricted to the pre-treatment period 1967–1969 because Salvador Allende took office in 1970. The main regressor is an interaction between an indicator for firms that were later nationalized and a linear time trend. The absence of statistically significant coefficients provides evidence consistent with parallel trends between nationalized and non-nationalized firms before Salvador Allende took office. All specifications include firm and sector-by-year fixed effects. Standard errors are clustered at the firm level. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A5: Other firm-level outcomes

	Log total assets		Leverage		Fixed over total assets		Log sales	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Ever nationalized × Allende period	-0.011	0.034	-0.005	0.001	0.032	0.019	-0.340	-0.287
	(0.131)	(0.140)	(0.031)	(0.034)	(0.041)	(0.042)	(0.205)	(0.222)
Observations	390	390	385	385	388	388	390	390
Firm fixed effects	Y	Y	Y	Y	Y	Y	Y	Y
Sector-year fixed effects	Y	Y	Y	Y	Y	Y	Y	Y
Covariates	N	Y	N	Y	N	Y	N	Y
Firms	71	71	71	71	71	71	71	71
Avg. dependent variable	16.70	16.70	0.386	0.386	0.660	0.660	15.74	15.74

Notes. The unit of observation is a firm-year. The table reports difference-in-differences estimates using alternative firm-level outcomes as dependent variables: log total assets in columns 1–2, leverage (debt over assets) in columns 3–4, fixed over total assets in columns 5–6, and log sales in columns 7–8. Odd-numbered columns include firm and sector-by-year fixed effects, while even-numbered columns additionally control for firm size in the pre-treatment period (1967–1969) interacted with year fixed effects. Standard errors are clustered at the firm level. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A6: Staggered difference-in-differences estimates

	ROA		Sales over assets		Export		Import	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Post takeover	-0.108***	-0.122***	-0.135***	-0.109**	0.049	0.069	-0.112	-0.160*
	(0.033)	(0.042)	(0.046)	(0.043)	(0.070)	(0.069)	(0.077)	(0.082)
Observations	390	390	390	390	389	389	389	389
Firm fixed effects	Y	Y	Y	Y	Y	Y	Y	Y
Sector-year fixed effects	Y	Y	Y	Y	Y	Y	Y	Y
Covariates	N	Y	N	Y	N	Y	N	Y
Firms	71	71	71	71	71	71	71	71
Avg. dependent variable	0.180	0.180	0.616	0.616	0.330	0.330	0.612	0.612

Notes. The unit of observation is a firm-year. The table reports staggered difference-in-differences estimates using the two-way fixed effects (TWFE) estimator. The dependent variables are return on assets (ROA) in columns 1–2, sales over assets in columns 3–4, an indicator equal to one if the firm exports in columns 5–6, and an indicator equal to one if the firm imports in columns 7–8. Odd-numbered columns include firm and sector-by-year fixed effects, while even-numbered columns additionally control for firm size in the pre-treatment period (1967–1969) interacted with year fixed effects. Standard errors are clustered at the firm level. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A7: Staggered difference-in-differences estimates (Borusyak et al., 2024)

	ROA	Sales over assets	Export	Import
	(1)	(2)	(3)	(4)
Panel A				
Post takeover	-0.119***	-0.209***	0.027	-0.113
	(0.042)	(0.054)	(0.067)	(0.069)
Panel B: Adding sector-year FE				
Post takeover	-0.103***	-0.150***	0.065	-0.153*
	(0.035)	(0.051)	(0.077)	(0.080)
Avg. dependent variable	0.180	0.643	0.326	0.612
p-value pre-trend (panel A)	0.312	0.100	0.182	0.252
p-value pre-trend (panel B)	0.220	0.136	0.300	0.081
Treated	32	32	32	32
Never treated	39	39	39	39

Notes. The unit of observation is a firm-year. The table reports staggered difference-in-differences estimates using the imputation-based estimator proposed by Borusyak et al. (2024). The dependent variables are return on assets (ROA) in column 1, sales over assets in column 2, an indicator equal to one if the firm exports in column 3, and an indicator equal to one if the firm imports in column 4. Panel A includes firm and year fixed effects, while panel B includes firm and sector-by-year fixed effects. Standard errors are clustered at the firm level. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A8: Robustness of changes in firm performance

	Collapse to two periods (before and during Allende)	Drops 1973 observations of December reporters	Replaces *sector* f.e. by narrower *industry* f.e.	Controls for the number of U.S. banks pre-Allende	Controls for probability (pscore) of being nationalized by Allende	Diff-in-diff + matching (Crump et al., 2009)	Diff-in-diff + matching (Yang and Ding, 2018)	Diff-in-diff + matching (Abadie, 2005)	Diff-in-diff + matching Doubly robust method
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A: Return on assets									
Ever nationalized × Allende period	-0.108***	-0.075**	-0.085**	-0.102***	-0.091***	-0.079***	-0.080***	-0.078***	-0.100***
	(0.034)	(0.032)	(0.034)	(0.032)	(0.034)	(0.028)	(0.028)	(0.024)	(0.030)
Avg. dependent variable	0.192	0.180	0.185	0.180	0.180	0.165	0.180	0.180	0.180
Panel B: Indicator importing									
Ever nationalized × Allende period	-0.240**	-0.214**	-0.237**	-0.208**	-0.258**	-0.203**	-0.204**	-0.187**	-0.213**
	(0.101)	(0.086)	(0.095)	(0.097)	(0.099)	(0.086)	(0.086)	(0.077)	(0.091)
Observations	118	370	370	389	389	362	389	389	389
Firms	59	71	67	71	71	65	71	71	71
Avg. dependent variable	0.637	0.612	0.625	0.612	0.612	0.628	0.612	0.612	0.612
Firm fixed effects	Y	Y	Y	Y	Y	Y	Y	Y	Y
Sector-year fixed effects	Y	Y	Y	Y	Y	Y	Y	Y	Y

Notes. The unit of observation is a firm-year in 1967–1973. The table reports robustness checks for the baseline difference-in-differences estimates using two measures of firm performance as dependent variables: return on assets in panel A and an indicator equal to one if the firm imports in panel B. Column 1 collapses the panel into two periods, before and during the Allende government. Column 2 excludes the 1973 observations of firms reporting in December because these reports include months following the September 11 military coup. The 1973 observations of firms reporting in July are retained because their fiscal years closed before the coup. The return-on-assets specification in column 2 contains 370 observations, while the importing specification contains 369 observations. Column 3 replaces sector-by-year fixed effects with more narrowly defined industry-by-year fixed effects. Column 4 controls for the number of U.S. banks in the pre-Allende period interacted with the Allende-period indicator. Columns 5–9 report estimates using alternative matching and reweighting procedures combined with the difference-in-differences design. Standard errors are clustered at the firm level. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A9: Controlling for pre-Allende international exposure

	Baseline importer interaction	Baseline exporter interaction	Both baseline-trade interactions
	(1)	(2)	(3)
Panel A: Return on assets			
Ever nationalized × Allende period	-0.108***	-0.101***	-0.112***
	(0.032)	(0.030)	(0.032)
Observations	390	390	390
Pre-Allende mean	0.180	0.180	0.180
Panel B: Sales over assets			
Ever nationalized × Allende period	-0.071	-0.107**	-0.085
	(0.057)	(0.047)	(0.056)
Observations	390	390	390
Avg. dependent variable	0.616	0.616	0.616
Panel C: Indicator exporting			
Ever nationalized × Allende period	-0.120	-0.019	-0.082
	(0.108)	(0.087)	(0.091)
Observations	389	389	389
Avg. dependent variable	0.330	0.330	0.330
Panel D: Indicator importing			
Ever nationalized × Allende period	-0.175*	-0.204**	-0.182**
	(0.091)	(0.094)	(0.091)
Observations	389	389	389
Avg. dependent variable	0.612	0.612	0.612
Importer in 1969 × Allende period	Y	N	Y
Exporter in 1969 × Allende period	N	Y	Y
Firm fixed effects	Y	Y	Y
Sector-year fixed effects	Y	Y	Y
Firms	71	71	71

Notes. The unit of observation is a firm-year. The table examines whether the baseline estimates are robust to allowing firms with different pre-Allende international exposure to experience distinct changes during the Allende period. Panel A uses return on assets as the dependent variable, panel B uses sales over assets, panel C uses an indicator equal to one if the firm exports, and panel D uses an indicator equal to one if the firm imports. Column 1 includes an interaction between an indicator for importing in 1969 and the Allende-period indicator. Column 2 analogously interacts exporting status in 1969 with the Allende-period indicator. Column 3 includes both interactions separately. The Allende-period indicator begins in 1970 for firms reporting in December and in 1971 for firms reporting in July. All specifications include firm and sector-year fixed effects. The reported dependent-variable means are calculated over the pre-Allende observations in the corresponding estimation samples. Standard errors are clustered at the firm level. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A10: Excluding comparison firms in industries with high nationalization intensity

	Exclude controls in the highest-intensity industry				Exclude controls in the three highest-intensity industries			
	Return on assets	Sales over assets	Indicator exporting	Indicator importing	Return on assets	Sales over assets	Indicator exporting	Indicator importing
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: Nationalization intensity measured by the share of firms								
Ever nationalized × Allende period	-0.057*	-0.101**	-0.045	-0.115	-0.057*	-0.098**	-0.036	-0.115
	(0.032)	(0.045)	(0.094)	(0.072)	(0.033)	(0.046)	(0.093)	(0.072)
Observations	334	334	333	333	326	326	325	325
R-squared	0.690	0.779	0.767	0.793	0.690	0.765	0.773	0.789
Firms	60	60	60	60	58	58	58	58
Avg. dependent variable	0.183	0.626	0.327	0.624	0.163	0.598	0.322	0.602
Panel B: Nationalization intensity measured by the share of pre-1970 assets								
Ever nationalized × Allende period	-0.081**	-0.103**	-0.084	-0.155**	-0.055	-0.095**	-0.063	-0.113
	(0.031)	(0.043)	(0.088)	(0.076)	(0.033)	(0.047)	(0.090)	(0.073)
Observations	381	381	380	380	321	321	320	320
Firms	69	69	69	69	57	57	57	57
Avg. dependent variable	0.161	0.591	0.324	0.591	0.161	0.596	0.331	0.581
Firm fixed effects	Y	Y	Y	Y	Y	Y	Y	Y
Year fixed effects	Y	Y	Y	Y	Y	Y	Y	Y

Notes. The unit of observation is a firm-year. The table examines whether the baseline results are sensitive to excluding potentially contaminated comparison firms operating in industries with high exposure to the UP nationalization program. The dependent variable is return on assets in columns 1 and 5, sales over assets in columns 2 and 6, an indicator equal to one if the firm exports in columns 3 and 7, and an indicator equal to one if the firm imports in columns 4 and 8. Industry-level nationalization intensity is calculated across 13 industries using a broader archival sample of 87 firms for which industry and nationalization status can be identified. This group consists of the 71 firms in the estimation panel and 16 additional firms excluded because they are observed only once or have missing data in key variables and therefore cannot contribute to specifications with firm fixed effects. In panel A, nationalization intensity is defined as the number of firms eventually nationalized divided by the total number of firms in the industry. In panel B, it is defined as the sum of average pre-1970 total assets among firms eventually nationalized divided by the corresponding sum among all firms in the industry. Assets are measured in levels, and the pre-1970 average is first calculated separately for each firm. Columns 1–4 exclude non-nationalized comparison firms operating in the single industry with the highest nationalization intensity. Columns 5–8 exclude non-nationalized comparison firms operating in the three industries with the highest nationalization intensity. Nationalized firms operating in these industries remain in the estimation sample. All specifications include firm and year fixed effects. Standard errors are clustered at the firm level. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A11: Heterogeneity by pre-existing state-bank relationships

	Z = State bank relationship in 1969	
Dependent variable:	ROA	Importing
	(1)	(2)
Ever nationalized × Allende period (β_1)	-0.205***	-0.272
	(0.035)	(0.254)
× Indicator for firm in Z (β_2)	0.156***	0.062
	(0.046)	(0.268)
Indicator for firm in Z × Allende period	-0.075**	0.121
	(0.036)	(0.075)
Observations	390	389
Firm fixed effects	Y	Y
Sector-year fixed effects	Y	Y
Firms	71	71
Firms ever nationalized and in Z	21	21
Avg. dependent variable	0.180	0.612
Effect for ever nationalized firms in Z ($\beta_1 + \beta_2$)	-0.049	-0.210
p-value ($\beta_1 + \beta_2$)	0.148	0.023

Notes. The unit of observation is a firm-year in the 1967–1973 period. The table reports difference-in-differences estimates allowing the effect of nationalization to vary with pre-existing financial linkages. Columns 1–2 interact the treatment with an indicator for firms that had a relationship with a state-owned bank in 1969. The row labeled "× Indicator for firm in Z" reports the interaction between the baseline treatment effect (Ever nationalized × Allende period) and the indicator for firms in category Z. The bottom rows report the implied treatment effect for nationalized firms in category Z. Standard errors clustered at the firm level are reported in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A12: Difference-in-differences estimates for banking relations

	Any international		Any U.S.		Any local private		Any state	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Ever nationalized × Allende period	0.009	0.018	0.026	0.083	0.128	0.158	-0.094	-0.025
	(0.108)	(0.121)	(0.098)	(0.106)	(0.091)	(0.097)	(0.084)	(0.086)
Observations	390	390	390	390	390	390	390	390
Firm fixed effects	Y	Y	Y	Y	Y	Y	Y	Y
Sector-year fixed effects	Y	Y	Y	Y	Y	Y	Y	Y
Covariates	N	Y	N	Y	N	Y	N	Y
Firms	71	71	71	71	71	71	71	71
Avg. dependent variable	0.628	0.628	0.493	0.493	0.585	0.585	0.841	0.841

Notes. The unit of observation is a firm-year. The table reports difference-in-differences estimates using measures of firms' banking relationships as dependent variables: an indicator equal to one if the firm has a relationship with an international bank in columns 1–2, an indicator equal to one if the firm has a relationship with a U.S. bank in columns 3–4, an indicator equal to one if the firm has a relationship with a domestic private bank in columns 5–6, and an indicator equal to one if the firm has a relationship with a state-owned bank in columns 7–8. Odd-numbered columns include firm and sector-by-year fixed effects, while even-numbered columns additionally control for firm size in the pre-treatment period (1967–1969) interacted with year fixed effects. Standard errors are clustered at the firm level. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A13: Observable sorting into public-firm employment

	Pre-Allende 1965–1969	Late Allende 1972–1973	Change in relationship
	(1)	(2)	(3)
Age	0.0050***	0.0018	
	(0.0008)	(0.0012)	
Age squared	-0.0001***	-0.0000	
	(0.0000)	(0.0000)	
Female	-0.0284***	-0.0384***	
	(0.0026)	(0.0059)	
Age × 1972–1973			-0.0032**
			(0.0015)
Age squared × 1972–1973			0.0000**
			(0.0000)
Female × 1972–1973			-0.0099
			(0.0064)
Workers	19,161	7,709	26,870
Share employed by public firms	0.039	0.056	0.044
R-squared	0.024	0.027	0.027
Education fixed effects	Y	Y	Y
Survey-year fixed effects	Y	Y	Y
p-value: demographic interactions			0.023
p-value: education interactions			0.419
p-value: all interactions			0.018

Notes. The unit of observation is a worker-survey observation. The dependent variable is an indicator equal to one if the worker's employer is explicitly classified by the *Encuesta de Ocupación y Desocupación* as a public firm and zero if it is classified as a private firm. The sample includes employed individuals aged 18–65. Columns 1 and 2 report linear probability models estimated separately for 1965–1969 and 1972–1973. Column 3 reports interaction coefficients from a pooled specification, where each worker characteristic is interacted with an indicator for 1972–1973. All specifications include education and survey-year fixed effects. The reported *p*-values test whether the demographic interactions, education interactions, and all interactions are jointly equal to zero. Standard errors are clustered at the sampling-segment level and reported in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A14: Public-private wage differences: Composition robustness

	Baseline	Activity-year fixed effects	Activity-year common support	Activity- and occupation-year fixed effects
	(1)	(2)	(3)	(4)
Panel A: Log wages				
Public firm × 1972–1973	-0.174***	-0.123***	-0.116***	-0.081*
	(0.035)	(0.041)	(0.041)	(0.042)
Observations	32,276	32,169	25,739	32,018
Pre-1972 mean	3.837	3.836	3.891	3.835
Panel B: Hours worked				
Public firm × 1972–1973	0.864	1.319	1.431*	1.238
	(0.614)	(0.803)	(0.794)	(0.810)
Observations	33,205	33,096	26,625	32,937
Pre-1972 mean	49.34	49.35	47.84	49.36
Panel C: Log hourly wages				
Public firm × 1972–1973	-0.182***	-0.143***	-0.139***	-0.099**
	(0.037)	(0.045)	(0.045)	(0.046)
Observations	31,817	31,711	25,391	31,561
Pre-1972 mean	0.010	0.009	0.088	0.007
Activity-year fixed effects	N	Y	Y	Y
Occupation-year fixed effects	N	N	N	Y
Activity-year common support	N	N	Y	N

Notes. The unit of observation is a worker-survey observation. The table reports changes in public-private wage, hours, and hourly-wage differentials during 1972–1973. The sample contains workers aged 18–65 employed in establishments explicitly classified by the survey as public or private firms. All specifications control for gender, age, age squared, and education. Column 1 includes survey-year fixed effects. Column 2 includes economic-activity-by-year fixed effects. Column 3 additionally restricts the sample to activity-year cells containing both public- and private-firm workers. Column 4 includes economic-activity-by-year and broad-occupation-by-year fixed effects. Economic activities and occupations classified as unspecified are excluded from the corresponding specifications. Heteroskedasticity-robust standard errors are reported in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table A15: Synthetic-control weights and diagnostic statistics

	U.S. imports from Chile	U.S. exports to Chile
	(1)	(2)
Panel A: Positive donor-country weights		
Argentina	—	0.137
Brazil	0.098	0.083
Colombia	0.193	—
Ecuador	0.246	0.120
Honduras	—	0.566
Mexico	—	0.095
Uruguay	0.406	—
Venezuela	0.058	—
Panel B: Fit and post-treatment gaps		
Donor countries	18	18
Pre-treatment years	15	15
Pre-treatment RMSPE	18.71	22.20
Pre-treatment RMSPE / pre-treatment mean (%)	9.54	11.05
Post/pre-treatment RMSPE	8.61	9.49
Mean observed trade, 1970–1973	107.50	239.75
Mean synthetic trade, 1970–1973	255.81	423.78
Mean gap, 1970–1973	-148.31	-184.03
Gap / synthetic mean (%)	-57.98	-43.43

Notes. This table reports details of the synthetic controls for U.S. imports from Chile and U.S. exports to Chile. The donor pool consists of 18 Latin American countries. Panel A reports countries receiving positive weight in at least one specification; omitted donor countries receive zero weight. Country weights are estimated separately for imports and exports, are constrained to be nonnegative, and sum to one before rounding. Each annual outcome from 1955 through 1969 is included as a predictor. RMSPE denotes the root mean squared prediction error between the observed and synthetic series. The post-treatment period is 1970–1973. The gap is defined as observed Chilean trade minus its synthetic counterpart. Trade-flow statistics are reported in the units of the underlying IMF *Direction of Trade Statistics* series.

Table A16: Nationalizations and voting

Dependent variable: Left-wing vote share			
Nationalizations:	*At least one nationalized plant*	*Number of nationalized plants*	*Nationalized plants over 1970 pop.*
	(1)	(2)	(3)
Nationalizations × 1973 Election	-4.02***	-0.36***	-1.75***
	(1.11)	(0.08)	(0.67)
Observations	2143	2143	2143
Municipality fixed effects	Y	Y	Y
Year fixed effects	Y	Y	Y
Avg. dependent variable	28.22	28.22	28.22
Municipalities	307	307	307

Notes. The unit of observation is a municipality-election. The table reports difference-in-differences estimates of the relationship between exposure to firm nationalizations and electoral support for the left-wing coalition. The sample includes 307 municipalities observed in four congressional elections (1961, 1965, 1969, and 1973) and three local elections (1963, 1967, and 1971). Left-wing vote share is defined as the vote share of Communist and Socialist candidates in each election. Exposure to nationalization is measured as an indicator for municipalities hosting at least one nationalized plant (column 1), the number of nationalized plants located in the municipality (column 2), and the number of nationalized plants per population, standardized by the mean and standard deviation (column 3). A total of 110 municipalities host at least one nationalized plant; conditional on exposure, the average number of nationalized plants is 3 (standard deviation 2). All specifications include municipality and election-year fixed effects. Standard errors, reported in parentheses, are clustered at the municipality level. Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

Table A17: Robustness of voting results

	Dependent variable: Left-wing vote share								
				Controlling by covariates × 1973					
	Congress elections	Collapse pre/post	*Variable* is linear pre-trend	Institutional presence	Spatial dependence	Socio-economic	*Variable* is non-nationalized large firms	Province-year fixed effects	Excl. province capitals
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Panel A: At least one nationalized plant									
Exposure to nationalization × 1973	-4.13***	-3.97***		-4.12***	-3.52***	-1.51	-3.67***	-2.73***	-5.42***
	(1.26)	(1.12)		(1.11)	(1.14)	(1.33)	(1.08)	(0.84)	(1.13)
Exposure to nationalization × *Variable*			0.29*				-0.75		
			(0.16)				(1.18)		
Panel B: Number of nationalized plants									
Exposure to nationalization × 1973	-0.34***	-0.36***		-0.40***	-0.34***	-0.18***	-0.75***	-0.27***	-1.66***
	(0.09)	(0.08)		(0.10)	(0.11)	(0.06)	(0.21)	(0.08)	(0.39)
Exposure to nationalization × *Variable*			0.01				0.43**		
			(0.01)				(0.20)		
Panel C: Nationalized plants over pop.									
Exposure to nationalization × 1973	-1.66**	-1.73**		-1.76**	-1.35**	-1.02**	-1.61***	-0.95*	-1.11
	(0.69)	(0.67)		(0.69)	(0.68)	(0.51)	(0.61)	(0.55)	(0.69)
Exposure to nationalization × *Variable*			0.03				-0.61		
			(0.08)				(0.49)		
Observations	1224	614	1529	2143	2143	2143	2143	2143	1968
Municipality fixed effects	Y	Y	Y	Y	Y	Y	Y	Y	Y
Year fixed effects	Y	Y	Y	Y	Y	Y	Y	Y	Y
Avg. dependent variable	27.01	30.99	24.81	28.22	28.22	28.22	28.22	28.22	28.33
Municipalities	307	307	307	307	307	307	307	307	282

Notes. This table presents robustness checks for the municipality-level voting results using difference-in-differences estimates. The sample includes 307 municipalities observed in four congressional elections (1961, 1965, 1969, and 1973) and three local elections (1963, 1967, and 1971). The cross-sectional treatment varies by panel: panel A uses an indicator for exposure to nationalization, panel B the number of nationalized plants, and panel C the number of nationalized plants per 1970 population. Each column implements a different robustness exercise: Column 1 restricts the sample to congressional elections; Column 2 collapses the data into pre- and post-1973 periods; Column 3 tests for pre-trends using a linear trend interacted with exposure; Columns 4–7 add alternative controls interacted with a 1973 indicator; Column 8 includes province-by-year fixed effects; and Column 9 excludes from the estimation the 25 municipalities that were province capitals in 1973. Standard errors in parentheses are clustered by municipality. Significance levels: *** p<0.01, ** p<0.05, * p<0.1.

Table A18: Other policies implemented by the UP government

Policy control:	Dependent variable: Left-wing vote share							
	Land reform	Trade protection	Milk distributed	Share mining workers	Share agricultural workers	Distance closest university	Illiteracy rate	Rural-urban migration
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Panel A: At least one nationalized plant								
Exposure to nationalization × 1973	-3.12***	-3.88***	-4.05***	-4.00***	-2.83**	-3.42***	-2.55**	-3.90***
	(1.09)	(1.17)	(1.11)	(1.11)	(1.14)	(1.27)	(1.10)	(1.09)
Policy × 1973	2.38***	-0.47	0.32	-1.89***	2.19***	0.36	1.76***	-1.47*
	(0.58)	(0.50)	(0.54)	(0.69)	(0.62)	(0.57)	(0.50)	(0.79)
Panel B: Number of nationalized plants								
Exposure to nationalization × 1973	-0.25***	-0.34***	-0.36***	-0.39***	-0.28***	-0.30***	-0.24***	-0.33***
	(0.07)	(0.07)	(0.08)	(0.09)	(0.06)	(0.10)	(0.06)	(0.08)
Policy × 1973	2.31***	-0.41	0.38	-2.24***	2.55***	0.46	1.80***	-1.09
	(0.60)	(0.44)	(0.55)	(0.72)	(0.67)	(0.57)	(0.58)	(0.76)
Panel C: Nationalized plants over pop.								
Exposure to nationalization × 1973	-1.37**	-1.71**	-1.74**	-1.83***	-1.27**	-1.30**	-1.11*	-1.72***
	(0.57)	(0.68)	(0.68)	(0.69)	(0.60)	(0.60)	(0.57)	(0.62)
Policy × 1973	2.58***	-0.64	0.19	-1.98***	2.91***	0.85*	2.03***	-1.62**
	(0.57)	(0.47)	(0.55)	(0.72)	(0.72)	(0.48)	(0.61)	(0.76)
Observations	2143	2143	2143	2143	2143	2143	2143	2143
Municipality fixed effects	Y	Y	Y	Y	Y	Y	Y	Y
Year fixed effects	Y	Y	Y	Y	Y	Y	Y	Y
Avg. dependent variable	28.22	28.22	28.22	28.22	28.22	28.22	28.22	28.22
Municipalities	307	307	307	307	307	307	307	307

Notes. The unit of observation is a municipality-election. Each column reports a difference-in-differences estimate using 307 municipalities observed in four congressional elections (1961, 1965, 1969, and 1973) and three local elections (1963, 1967, and 1971). Columns differ only in the inclusion of an alternative policy exposure variable, each interacted with an indicator for the 1973 election, to account for the local impact of major contemporaneous policies implemented between 1970 and 1973. All policy variables are standardized. Column 1 controls for land reform (share of land expropriated), column 2 for trade protection (tariffs interacted with local agricultural production), column 3 for exposure to the milk program, columns 4 and 5 for employment shares in mining and agriculture, column 6 for distance to the nearest university, column 7 for local illiteracy rates, and column 8 for changes in rurality between 1960 and 1970. All specifications include municipality and election-year fixed effects. Standard errors are clustered at the municipality level. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.